

The School of Mathematics



THE UNIVERSITY
of EDINBURGH

Fairness-Constrained Bayesian Search for Optimal Volt–Watt Settings in Low-Voltage Distribution Networks

by

Ruoxi Wang

Dissertation Presented for the Degree of
MSc in Statistics and Operational Research

August 2026

Supervised by
Dr Miguel F. Anjos

Abstract

The increasing integration of distributed energy resources (DERs) in low-voltage distribution networks can create local voltage constraints and unequal active power curtailment among DERs at different electrical locations. This dissertation aims to determine DER-specific Volt–Watt parameters that maximise total DER production while limiting disparity in DER utilisation.

A LinDistFlow-based optimal power flow (OPF) framework is developed to optimise these parameters across multiple operating scenarios. Fairness is introduced through pairwise fairness-stick coefficients that coordinate the relative positions of the Volt–Watt characteristics. Grid search and Bayesian optimisation are compared in terms of solution quality and computational performance, and the framework is extended to larger DER configurations to assess search scalability. The LinDistFlow formulation is also evaluated against a second-order cone branch-flow model.

The results show that maximising total DER production without fairness constraints can produce differences in DER utilisation. The fairness-stick formulation reduces this disparity while retaining more than 97% of the unconstrained production in the original three-DER case. For this case, both search methods identify solutions of comparable quality, while Bayesian optimisation requires less than half as many OPF evaluations and less than half the computation time. As more DERs are introduced, the production reduction associated with fairness increases and the parameter-search problem becomes higher-dimensional. Within the prescribed evaluation budgets, Bayesian optimisation continues to identify fairness-feasible solutions in the extended cases, whereas grid search becomes limited by increasingly sparse coverage of the search space.

Overall, the study demonstrates the trade-off between DER production and utilisation fairness and highlights the value of efficient parameter-search methods as coordinated Volt–Watt control is extended to larger numbers of DERs.

Acknowledgments

I would like to express my sincere gratitude to my supervisor, Dr Miguel F. Anjos, for his continued guidance and support throughout this project. His insight, encouragement and willingness to help at every stage were invaluable in shaping both the direction of this research and my development.

I would also like to thank Mihajlo Curcic of Hydro-Québec for his mentorship throughout the course of this project. His patience and generosity with his time were greatly appreciated, and his guidance extended well beyond technical support to include broader discussions on the direction and objectives of the research. This work would not have been possible without his input.

I am deeply grateful to my parents for their unwavering support and encouragement throughout my studies. Their belief in me has been a constant source of motivation.

Finally, I would like to acknowledge my own perseverance and commitment throughout this project. Completing this thesis involved sustained effort over an extended period, making its completion particularly meaningful to me.

Own Work Declaration

I declare that this thesis is my own work and that all material presented is original, except where otherwise clearly acknowledged and appropriately referenced.

Contents

1	Introduction	1
1.1	Background	1
1.2	Research Objectives and Contributions	1
1.3	Structure of the Dissertation	2
2	Literature Review	3
2.1	Low-Voltage Distribution Networks and Voltage Regulation	3
2.2	Volt–Watt Control	4
2.3	Fairness in Volt–Watt Control	4
2.4	OPF-Based Network-Constrained Volt–Watt Optimisation	5
2.5	Volt–Watt Parameter Optimisation	6
2.6	Chapter Summary	7
3	Methodology	8
3.1	Overview of the Proposed Methodology	8
3.2	Optimal Power Flow Formulation	8
3.2.1	Distribution Network Model	8
3.2.2	OPF Formulation	9
3.2.3	Model Assumptions and Limitations	10
3.3	Local Volt–Watt Control Formulation	10
3.3.1	Volt–Watt Constraint	10
3.3.2	DER-Specific Volt–Watt Shift Parameters	11
3.4	Fairness-Constrained Formulation	11
3.5	Optimisation Methods	12
3.5.1	Grid Search	13
3.5.2	Bayesian Optimisation	13
3.5.3	Gaussian Process Surrogate Model	14
3.6	Probabilistic Operating Scenario Generation	14
3.7	Model Validation Using an SOC Formulation	14
3.8	Case Study Configuration	15
3.9	Chapter Summary	16
4	Results	17
4.1	Overview	17
4.2	Baseline Volt–Watt Optimisation	17
4.2.1	Optimised Parameters and Network Performance	17
4.2.2	DER Utilisation and Curtailment	18
4.3	Fairness-Constrained Volt–Watt Optimisation	19
4.3.1	Optimised Parameters and Network Performance	19
4.3.2	DER Utilisation and Curtailment	21
4.3.3	Production–Fairness Trade-off	23
4.4	Comparison of Search Methods	24
4.4.1	Search Efficiency and Solution Quality	24
4.4.2	Convergence Behaviour	24
4.5	Computational Scalability with Increasing Numbers of DERs	25
4.5.1	Motivation	25
4.5.2	Optimisation Performance	25
4.5.3	Computational Performance	27
4.6	SOC Validation	28
4.6.1	Voltage Accuracy	28
4.6.2	Feasibility and Relaxation Tightness	29
4.7	Chapter Summary	30

5	Conclusions and Future Work	31
5.1	Conclusions	31
5.2	Limitations and Future Work	31

List of Tables

1	Case study configuration for the baseline three-DER formulation	16
2	Optimised Volt–Watt shift voltages obtained for the unconstrained formulation.	17
3	Overall network performance under the unconstrained formulation.	18
4	DER performance under the unconstrained formulation.	18
5	Optimised fairness-stick coefficients and resulting Volt–Watt shift voltages.	20
6	Overall network performance under the fairness-constrained formulations.	21
7	Comparison of average DER utilisation (%).	21
8	Comparison of cumulative DER curtailment.	22
9	Comparison of search efficiency and solution quality.	24
10	Comparison of optimisation performance for increasing numbers of participating DERs.	26
11	Reduction in total DER production associated with the fairness constraint.	26
12	Comparison of total computation time for increasing numbers of participating DERs.	27
13	Retrospective identification of the final Bayesian-optimisation solution within the prescribed evaluation budget.	27
14	Voltage accuracy obtained from SOC validation.	28
15	SOC feasibility and relaxation tightness.	29

List of Figures

1	Topology of the modified 13-bus radial low-voltage distribution feeder. Bus 0 is the slack bus, and DERs are connected at Buses 3, 6 and 12.	8
2	Optimised Volt–Watt characteristics and operating points for the unconstrained formulation.	18
3	Optimised Volt–Watt characteristics under the fairness-constrained formulation: (a) grid search and (b) Bayesian optimisation.	20
4	Comparison of average DER utilisation under the unconstrained and fairness-constrained formulations.	22
5	Comparison of cumulative DER curtailment under the unconstrained and fairness-constrained formulations.	23
6	Relationship between total DER production and average utilisation disparity.	23
7	Convergence histories of grid search and Bayesian optimisation.	25

1 Introduction

1.1 Background

The global electricity sector is undergoing a transition towards low-carbon energy systems, driven by climate policy commitments, declining renewable energy costs and increasing demand for sustainable electricity. Solar photovoltaic (PV) generation has become a major contributor to new electricity generation capacity worldwide. A substantial share of this expansion has occurred at the distribution level, where residential and commercial rooftop PV systems are increasingly connected directly to low-voltage distribution networks [21].

The growth of distributed PV is changing the operation of electricity distribution systems. Distribution networks have traditionally been designed to deliver electricity from centralised power stations to passive consumers under relatively predictable operating conditions. As a growing share of customers become both electricity consumers and producers, local generation has become an increasingly important component of feeder power flows. The simultaneous electrification of transport and heating is further increasing the diversity of electricity demand and placing additional operational requirements on existing distribution infrastructure [10].

Maintaining secure network operation under these conditions has become an important challenge for distribution system operators. High levels of distributed PV can lead to voltage regulation problems, renewable energy curtailment and reduced utilisation of existing network capacity if appropriate control strategies are not adopted [25]. Traditional solutions such as network reinforcement and additional voltage regulation equipment remain available, but they generally require significant investment and long implementation times. Control approaches that make use of the flexibility available within distributed energy resources (DERs) provide another means of managing increasing PV penetration.

The development of smart photovoltaic inverters has provided additional options for voltage regulation without extensive infrastructure upgrades. Volt–Watt control is one of the grid-support functions available in modern inverters for mitigating voltage rise associated with distributed PV generation [19]. Its use provides an alternative to relying solely on network-level voltage regulation measures as PV penetration increases.

The performance of Volt–Watt control is sensitive to the choice of its control parameters, which can lead to different levels of photovoltaic utilisation, voltage regulation and active power curtailment under the same network conditions. Suitable settings may also vary across a feeder because individual photovoltaic systems experience different electrical conditions depending on their location. Determining appropriate Volt–Watt parameters therefore requires network operating constraints, photovoltaic utilisation and the distribution of curtailment among participating systems to be considered together. The computational cost of evaluating candidate settings adds a further consideration when the parameter space becomes larger.

The DERs considered in this dissertation are solar photovoltaic (PV) systems. This dissertation addresses these considerations through an optimal power flow (OPF)-based optimisation framework for Volt–Watt parameter selection, incorporating an explicit fairness constraint and computationally efficient parameter search.

1.2 Research Objectives and Contributions

The overall aim of this dissertation is to determine Volt–Watt control parameters that maximise total DER production in a low-voltage distribution network while limiting the disparity in utilisation among participating DERs. The specific objectives are to:

1. Develop an OPF-based model that represents the relationship between Volt–Watt parameters, DER production and network operating limits, and validate the linearised formulation against a second-order cone (SOC) branch-flow model.
2. Develop a fairness-stick formulation that coordinates the relative Volt–Watt parameters of participating DERs, and apply Bayesian optimisation to efficiently search the resulting fairness-constrained parameter space under a limited number of OPF evaluations.

3. Evaluate Bayesian optimisation against a grid search baseline using the same OPF model and operating scenarios, with comparison based on total DER production, fairness and computational efficiency, including the effect of increasing the number of participating DERs on search performance.

This dissertation combines fairness-aware Volt–Watt parameter selection with an OPF-based network evaluation framework, and provides an empirical comparison of grid search and Bayesian optimisation under the same network conditions, considering both the quality of the resulting parameter settings and their computational requirements. The comparison is extended to higher-dimensional DER configurations to assess how the performance of the two search methods changes as the optimisation problem grows.

1.3 Structure of the Dissertation

The remainder of this dissertation is organised as follows. Chapter 2 reviews the relevant literature on voltage regulation in low-voltage distribution networks, Volt–Watt control, fairness in active power curtailment, and existing approaches to Volt–Watt parameter optimisation, concluding with a summary that motivates the fairness-constrained optimisation framework developed in this dissertation. Chapter 3 presents the proposed methodology, including the OPF-based network model, the fairness-constrained Bayesian optimisation framework, and the SOC branch-flow formulation used to validate the underlying network model. Chapter 4 presents the results obtained using the proposed approach, compared against a grid search baseline, together with the corresponding SOC validation, and examines how this comparison scales as the number of participating DERs is increased from three to six. Chapter 5 concludes the dissertation, summarising its key findings and discussing the limitations of the proposed framework and directions for future work.

2 Literature Review

2.1 Low-Voltage Distribution Networks and Voltage Regulation

Low-voltage (LV) distribution networks form the final stage of the electricity supply system, delivering electricity from distribution transformers to residential, commercial and small industrial customers [1]. These networks are predominantly configured as radial feeders [6], originally designed for unidirectional power transfer from the upstream grid to downstream loads, with each customer supplied through a single electrical path.

Compared with transmission systems, LV distribution networks serve a large number of customer connection points. Their operation is affected by variations in customer demand and the increasing presence of bidirectional power flows associated with distributed generation [1].

The increasing deployment of distributed photovoltaic (PV) generation has changed the traditional operation of LV distribution networks. Many customers now generate electricity locally and export surplus power back into the network, and feeder operating conditions become more variable throughout the day than under conventional passive operation. At high levels of PV penetration, this exported power can produce reverse power flow along distribution feeders and contribute to voltage rise. Voltage regulation has therefore become an important consideration for accommodating increasing levels of PV generation in LV networks [27].

Voltage rise most commonly occurs when high photovoltaic generation coincides with low local electricity demand. Under these conditions, surplus active power exported from customers towards the upstream network can push feeder voltage upward. Instead of the gradual voltage drop associated with conventional downstream power flow, feeder voltage may rise and eventually exceed the permissible operating limit [27].

The severity of voltage rise is related to the electrical characteristics of the feeder. LV distribution networks typically exhibit high resistance-to-reactance ratios, making bus voltages particularly sensitive to active power injections. Electrical location also plays an important role because feeder impedance accumulates with distance from the transformer. Customers located near the end of the feeder generally experience larger voltage rises than those closer to the transformer [11]. Excessive voltage rise limits the amount of photovoltaic generation that can be accommodated by the network. Once the upper voltage limit is reached, additional PV generation cannot be exported without corrective action, reducing the available hosting capacity of the feeder [27].

Several approaches have been developed to regulate voltage in low-voltage distribution networks as photovoltaic penetration has increased. Conventional network-based solutions include on-load tap changers, voltage regulators, capacitor banks and feeder reinforcement. These methods regulate feeder voltage by modifying network operating conditions or increasing the electrical capacity of the distribution system [27]. Although these approaches remain effective, they have traditionally been applied in networks where voltage variations were primarily driven by changes in customer demand. Fluctuations in photovoltaic output can produce voltage variations over shorter time scales, creating additional challenges for conventional voltage regulation equipment. Frequent operation of such equipment may also increase maintenance requirements and reduce equipment lifetime [24].

Network reinforcement provides a direct means of increasing voltage headroom by reducing feeder impedance. Larger conductors and upgraded network assets can accommodate higher levels of distributed generation, but these solutions require capital investment and are generally implemented over long planning timescales. Modern photovoltaic inverters provide an alternative means of supporting voltage regulation within existing distribution infrastructure because their power output can be actively controlled in response to network conditions.

Reactive power support has been widely investigated because inverter-generated reactive power can influence local voltage without requiring major network upgrades [25]. Its ability to mitigate voltage rise is subject to practical limitations. The reactive power capability of a PV inverter is constrained by its apparent power rating, which may limit its availability when active power output is high [31]. The effectiveness of reactive power control is also affected by the resistive characteristics of LV feeders, which can limit the voltage response obtained from reactive power adjustment [34].

2.2 Volt–Watt Control

The increasing penetration of photovoltaic generation has stimulated the development of inverter-based voltage regulation strategies that make use of the control capability of modern grid-connected photovoltaic inverters. Unlike conventional inverters that convert direct-current electricity into alternating current, modern photovoltaic inverters are capable of adjusting their power output in response to measured grid conditions. IEEE Standard 1547-2018 [19] defines voltage-active power control, commonly referred to as Volt–Watt control, as one of the standard grid-support functions that distributed energy resources (DERs) can implement to assist voltage regulation.

Volt–Watt control regulates the active power exported by a photovoltaic inverter according to the voltage measured at its point of connection. Under normal operating conditions, the inverter exports all available photovoltaic power. Once the local voltage exceeds a predefined threshold, the inverter progressively reduces its active power output following a prescribed Volt–Watt characteristic. The amount of power curtailment increases as the measured voltage rises, allowing feeder voltages to remain within acceptable operating limits and reducing the need for complete inverter disconnection [12]. Depending on feeder sensitivity, voltage may remain above the permissible limit even when reactive power output has reached its maximum, in which case active power curtailment becomes necessary to restore voltage within bounds [17].

Volt–Watt control has several practical characteristics that support its application in photovoltaic-rich distribution networks. It can operate using voltage measurements available locally at the inverter, avoiding the need for continuous communication with a central controller. This makes the method well suited to feeders with large numbers of geographically dispersed photovoltaic systems, since each inverter can regulate its own output independently of the others. The performance of Volt–Watt control depends on the selection of its control parameters. Fixed settings designed for one operating condition can become unnecessarily conservative or insufficiently effective as feeder loading, upstream voltage and photovoltaic generation change over time. Uniform Volt–Watt settings can also produce location-dependent differences in curtailment among participating PV systems [23].

Optimisation algorithms provide a means of selecting Volt–Watt voltage thresholds and control settings according to feeder conditions instead of applying the same predefined curve to every inverter. Individually selected settings have been shown to achieve higher photovoltaic utilisation than default parameters while satisfying network operating constraints [2]. The selected parameters can also affect the distribution of active power curtailment among photovoltaic systems, introducing an additional consideration in Volt–Watt parameter selection.

2.3 Fairness in Volt–Watt Control

Fairness in Volt–Watt control is generally understood as the extent to which curtailment is distributed equally among participating photovoltaic systems, rather than concentrated on a subset of customers [29]. Unlike voltage regulation, which can be assessed directly from measured bus voltages, fairness has been interpreted using a variety of metrics in the literature, with no single measure adopted universally. Meeting voltage limits alone does not determine whether the resulting allocation of curtailment is fair [26].

Fairness can be defined according to different operational outcomes. Equal PV harvesting, equal energy export and equal financial benefit have each been proposed as fairness criteria for active power curtailment schemes [26]. These criteria can produce different outcomes because a household’s exported energy or realised financial benefit depends on its own demand pattern relative to PV generation as well as the curtailment applied. Conventional Volt–Watt control determines PV curtailment from local voltage, which can place a greater curtailment burden on households located farther from the distribution transformer [26, 16]. The selection of a fairness criterion therefore depends on the outcome that the curtailment scheme is intended to achieve.

Among these alternative interpretations, this dissertation adopts equal DER utilisation as its operational definition of fairness. This measure reflects the proportion of available generation utilised by each DER without incorporating household-specific demand or tariff structures. The fairness objective adopted in this dissertation is to maximise total DER production while keeping the disparity

in DER utilisation across participating units within an acceptable limit.

A range of coordinated Volt–Watt strategies has been proposed to reduce unequal curtailment. Early work assigned different droop coefficients to individual photovoltaic systems so that curtailment could be distributed across the feeder while voltages remained within statutory limits [33]. Common or coordinated curtailment factors have also been calculated and applied across photovoltaic inverters through distributed or centralised computation. This approach can reduce the dependence of curtailment on inverter location and improve the distribution of active power reduction among customers [15]. Voltage sensitivity information has also been incorporated into controller design, allowing curtailment decisions to account for the contribution of each photovoltaic system to local voltage rise [8].

Recent studies have considered differences in the operating conditions experienced by customers at different locations along the feeder. Fairness can be evaluated among photovoltaic systems with similar voltage sensitivities or exposure to the same network constraints instead of evaluating all systems as a single group. A study introduced the concept of fairness domains, in which photovoltaic systems are grouped according to their electrical interactions and voltage sensitivity [22]. Within each domain, curtailment is allocated among participating systems, while different domains respond according to their available voltage headroom. Domain-level fairness has been shown to improve annual photovoltaic energy while maintaining voltage security. It can also avoid conclusions based on a single global fairness index that does not account for differences in the network conditions experienced by individual photovoltaic systems [22].

The methods considered in the literature use different mechanisms to distribute active power curtailment among photovoltaic systems while maintaining network operating requirements. Their performance can depend on the representation of each PV system’s voltage sensitivity and the underlying relationship between voltage and power injection. The choice of fairness formulation also varies according to the network conditions and the quantity used to evaluate equitable curtailment.

2.4 OPF-Based Network-Constrained Volt–Watt Optimisation

Optimal Power Flow (OPF) provides a framework for determining Volt–Watt parameters while accounting for the physical behaviour of the distribution network and the operational limits of distributed energy resources.

In an OPF formulation, the electrical relationships governing power flows are combined with constraints such as voltage limits, branch power flows and inverter operating limits [14]. When applied to Volt–Watt parameter optimisation, this framework allows the network effects of different parameter settings to be evaluated while accounting for these operating constraints.

Early studies generally used OPF to calculate the optimal active and reactive power outputs of photovoltaic inverters under a given operating condition. The resulting power set-points were then communicated directly to individual inverters for implementation [9]. This form of control can improve voltage regulation and reduce network losses, but changes in network conditions require new operating points to be calculated and dispatched. The required computation and communication increase as more distributed photovoltaic systems are connected and operating conditions change more frequently.

Later work considered the parameters of the Volt–Watt control curve as optimisation variables instead of directly determining inverter power outputs. The optimisation determines the parameters of the local controller, after which each inverter regulates its active power output according to locally measured voltage. Once the parameters have been determined, changes in operating conditions are handled through the specified Volt–Watt characteristic without repeated dispatch of active power set-points. This retains network-level optimisation during parameter selection and allows the resulting control to operate locally at each inverter [30].

Several optimisation strategies have been proposed for determining Volt–Watt parameters. One approach formulates the active and reactive power allocation of each smart inverter as a mathematical optimisation problem and subsequently derives Volt–Var and Volt–Watt control curves from the resulting optimal operating points. Distribution network configuration, operating objectives, solar irradiation and load conditions are included as optimisation inputs, and the resulting operating points are used to determine the control curves [35]. Another approach incorporates Volt–Var and Volt–Watt

droop parameters directly into an OPF formulation. These parameters correspond to the threshold and slope settings defining the control characteristics introduced in Section 2.2 and are treated explicitly as optimisation variables [28]. This formulation allows the droop settings to be determined while considering network-level objectives such as voltage regulation and network loss minimisation.

Voltage-sensitivity-based methods provide an alternative to optimisation-based parameter selection. One approach determines Volt–VAR–Watt parameters directly from the network impedance matrix [36]. Since the voltage sensitivity matrix depends on feeder electrical characteristics, the parameter settings can be calculated without repeated power-flow calculations or communication among inverters. The reported results showed voltage-control performance comparable to optimisation-based approaches with lower computational complexity [36]. Parameter selection in this case is determined analytically from voltage sensitivities, which provides less flexibility for incorporating several network objectives within the same formulation. OPF provides greater flexibility when the parameter selection must account for multiple operational objectives and network constraints.

OPF-based studies use different optimisation objectives according to the intended application, including voltage regulation, network loss minimisation, active power curtailment minimisation and improved photovoltaic utilisation. Fairness has also been incorporated into OPF formulations that directly determine curtailment set-points [26]. The choice of objective therefore determines how the available network capacity is allocated between competing performance criteria.

Incorporating droop settings directly as constraints within a distribution grid optimal power flow formulation can make the resulting model non-convex and more computationally difficult to solve than a standard convex OPF [20]. The droop relationship is piecewise linear, and its exact representation within an OPF formulation may require integer variables to identify the active segment of the control curve at each operating point. This produces a mixed-integer formulation rather than a continuous convex problem. Continuous reformulations have consequently been developed to represent droop behaviour without discrete variables [20].

An alternative is to hold parameters that coordinate the relative settings of different distributed energy resources fixed during each OPF solution, while allowing the underlying Volt–Watt shift parameters themselves to remain optimisation variables within the OPF. The OPF then evaluates network performance for a given coordination setting, and the coordination parameters are determined through an external search process using the OPF evaluations.

2.5 Volt–Watt Parameter Optimisation

External parameter search can require a large number of OPF evaluations as the number of candidate settings increases. Different parameter combinations can produce different levels of photovoltaic utilisation, voltage regulation and active power curtailment. The computational requirement grows with the number of externally searched parameters because each candidate combination requires one or more complete OPF evaluations, which may involve a joint multi-scenario optimisation problem.

Grid search is a simple approach for parameter optimisation. The search space is discretised into a finite set of candidate values, and every possible combination is evaluated explicitly [5]. This makes the method straightforward to implement and reproduce, while allowing the best solution within the predefined grid to be identified.

The computational cost of grid search increases with the number of decision variables and the resolution of the search grid. The number of candidate combinations grows exponentially with the dimensionality of the search space, a phenomenon commonly referred to as the curse of dimensionality [4]. Exhaustive evaluation can therefore become computationally demanding when the search space is large or when each objective evaluation requires substantial computation.

Alternative optimisation methods include random search [4] and population-based approaches such as genetic algorithms and evolutionary strategies [5]. These methods do not require evaluation of every point on a predefined grid and can reduce the number of candidate solutions considered. Evolutionary strategies can explore the search space more selectively than exhaustive grid search, although they may still require many objective evaluations before reaching a suitable solution. This becomes relevant when each evaluation requires the solution of an OPF problem.

Bayesian optimisation is designed for optimisation problems in which objective-function evaluations

are expensive. A probabilistic surrogate model is constructed from previously evaluated parameter settings and their corresponding objective values. An acquisition function uses this model to select the next parameter set for evaluation, accounting for both unexplored regions of the search space and regions associated with favourable predicted objective values. The surrogate model is updated after each new evaluation, allowing subsequent parameter selections to use information obtained from earlier iterations. This procedure can identify high-quality solutions with fewer objective-function evaluations than exhaustive search [7].

These characteristics are particularly well suited to OPF-based Volt–Watt parameter optimisation. As noted above, each candidate parameter set must be evaluated through a constrained OPF model solved under multiple operating scenarios, making Bayesian optimisation especially attractive for problems combining expensive black-box objective functions with limited computational budgets [32].

Bayesian optimisation has also been extended to problems involving explicit constraints. Constrained Bayesian optimisation represents information about both the objective and feasibility conditions during the search, allowing candidate points to be selected according to their predicted performance and likelihood of satisfying the specified constraints [18]. This formulation is applicable to engineering optimisation problems in which feasible parameter settings must satisfy operational requirements.

In the baseline three-DER case, grid search and Bayesian optimisation are applied to three externally searched coefficients that coordinate the relative Volt–Watt shift parameters of the participating DERs. The same search framework is subsequently extended to higher-dimensional DER configurations. Grid search provides a systematic benchmark for evaluating the predefined coefficient space, while Bayesian optimisation provides an alternative strategy that uses information from previous evaluations to guide subsequent parameter selection. The two methods are compared in terms of solution quality and computational requirements.

2.6 Chapter Summary

The literature shows that increasing PV penetration creates voltage regulation challenges in low-voltage networks and that inverter-based active power control provides a means of managing these conditions without relying solely on conventional network assets. The effectiveness of Volt–Watt control depends on its parameter settings, which influence DER utilisation and the amount of active power curtailment.

The studies reviewed in this chapter also show that parameter selection must account for the electrical constraints of the feeder. OPF provides a framework for representing these constraints, while different parameter-search methods offer alternative ways of managing the computational effort associated with repeated network evaluations. Fairness introduces an additional consideration because control settings that satisfy voltage requirements can still produce different utilisation outcomes among DERs.

These considerations provide the basis for the fairness-constrained Volt–Watt optimisation framework developed in the following chapter.

3 Methodology

3.1 Overview of the Proposed Methodology

This chapter develops a methodology for determining local Volt–Watt control parameters for distributed energy resources (DERs) connected to a low-voltage feeder. The objective is to maximise total DER production while satisfying network operating constraints and achieving an acceptable level of fairness between DERs connected at different locations along the feeder.

The proposed optimisation framework uses probabilistic scenario generation to represent variability in customer demand and upstream network conditions. For the resulting operating scenarios, an optimal power flow (OPF) model determines the Volt–Watt shift parameters and evaluates DER production jointly across all scenarios. For the baseline three-DER case, fairness is introduced through three pairwise coefficients that constrain the relative positions of the DER-specific Volt–Watt characteristics. Since each candidate coefficient set requires a complete multi-scenario OPF evaluation, both a coarse-to-fine grid search and Bayesian optimisation based on a Gaussian process surrogate model are used to search the coefficient space, and their performance is subsequently compared.

3.2 Optimal Power Flow Formulation

3.2.1 Distribution Network Model

The proposed methodology is implemented on a modified 13-bus radial low-voltage distribution feeder. The network has 13 buses connected by 12 branches, with Bus 0 as the slack bus. Let $\mathcal{N}$ denote the set of all buses in the network. Three distributed energy resources (DERs) are connected at buses 3, 6 and 12, while the remaining buses carry passive customer loads. In this dissertation, the terms bus and node are used interchangeably to denote connection points in the distribution network.

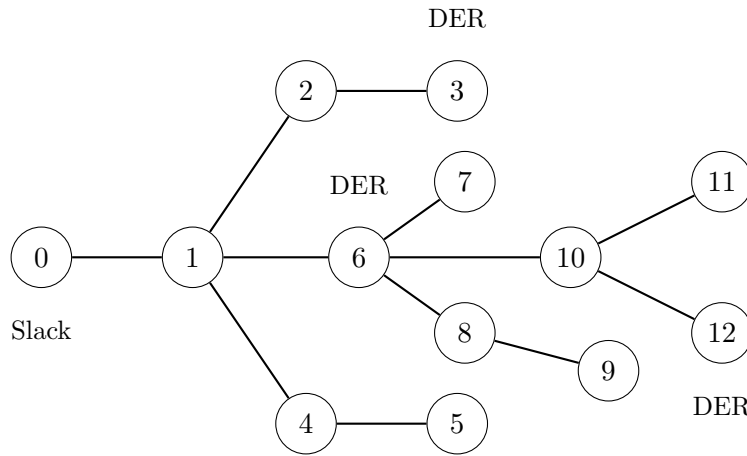


Figure 1: Topology of the modified 13-bus radial low-voltage distribution feeder. Bus 0 is the slack bus, and DERs are connected at Buses 3, 6 and 12.

Each branch (i, j) is characterised by a resistance r_{ij} and reactance x_{ij} . In the original feeder data, resistance does not dominate reactance, whereas low-voltage distribution feeders typically exhibit a relatively high resistance-to-reactance ratio. A more resistive network is particularly relevant to the present study because active power injections then have a stronger influence on local voltage, making Volt–Watt control suitable for mitigating voltage rise.

To represent these characteristics, the original resistance and reactance values are swapped and rescaled before use.

$$r_{ij} = 0.075 x_{ij}^{\text{orig}}, \quad x_{ij} = 0.075 r_{ij}^{\text{orig}}, \quad (3.1)$$

where r_{ij}^{orig} and x_{ij}^{orig} are the original branch resistance and reactance values. This modification produces a more resistive feeder representation, and the resulting parameters are used throughout the

remainder of this study.

The feeder topology is represented using parent–child relationships. Each non-slack bus has exactly one upstream parent bus and zero or more downstream child buses, defining the radial structure of the network.

Power flows and bus voltages are represented using the LinDistFlow approximation [3]. The voltage relationship therefore accounts for both active and reactive power flows through the corresponding resistance and reactance. The resulting linear formulation avoids the nonlinear loss terms of the full DistFlow equations, allowing the OPF to be solved repeatedly across the operating scenarios and candidate Volt–Watt settings considered in the optimisation procedure.

3.2.2 OPF Formulation

The OPF model determines the DER-specific Volt–Watt shift parameters together with the corresponding network operating variables across all operating scenarios. The optimisation maximises total DER production while satisfying the prescribed network, generation and voltage constraints.

The OPF is solved jointly across all operating scenarios so that each DER’s Volt–Watt parameter is evaluated consistently under the full scenario set.

The network operating variables include branch active and reactive power flows $P_{ij,s}$ and $Q_{ij,s}$, squared bus voltages $v_{i,s}$, and DER active power output $p_{i,s}^{\text{DER}}$. These variables are defined separately for each operating scenario $s \in \Omega_S$. DER active power output is defined only at the DER buses $\Omega_{\text{DER}} = \{3, 6, 12\}$. A Volt–Watt shift parameter is assigned to each DER and shared across all operating scenarios.

The objective is to maximise total DER power output across all DERs and operating scenarios:

$$\max \sum_{s \in \Omega_S} \sum_{i \in \Omega_{\text{DER}}} p_{i,s}^{\text{DER}}. \quad (3.2)$$

Throughout this dissertation, this summed quantity is referred to as total DER production for conciseness. Since the operating scenarios represent independent snapshots rather than sequential time periods, it should be interpreted as the sum of DER power output across the scenario set rather than as a cumulative energy quantity. With equally weighted scenarios, maximising this quantity is equivalent to maximising the average DER power output across the scenarios.

This objective is maximised subject to the network constraints described below. For each non-slack bus j , let $i = \text{parent}(j)$ denote its upstream bus. The active power balance is

$$P_{ij,s} = p_{j,s}^L - p_{j,s}^{\text{DER}} + \sum_{k \in \mathcal{C}(j)} P_{jk,s}, \quad (3.3)$$

where $p_{j,s}^L$ is the active power demand at bus j , and $\mathcal{C}(j)$ denotes the set of buses directly downstream of bus j . The power entering bus j from its upstream branch supplies the local active demand and downstream branch flows after accounting for any active power generated by a DER connected at that bus.

The corresponding reactive power balance is

$$Q_{ij,s} = q_{j,s}^L + \sum_{k \in \mathcal{C}(j)} Q_{jk,s}, \quad (3.4)$$

where $q_{j,s}^L$ is the reactive power demand at bus j . Reactive power support from the DERs is not included, so reactive branch flows are determined by customer reactive demand and downstream reactive power requirements.

Bus voltages are represented using squared voltage magnitudes. The voltage relationship across each branch follows the LinDistFlow formulation,

$$v_{j,s} = v_{i,s} - 2(r_{ij}P_{ij,s} + x_{ij}Q_{ij,s}), \quad (3.5)$$

where $v_{i,s}$ denotes the squared voltage magnitude at bus i in scenario s .

The slack-bus voltage is fixed separately for each scenario,

$$v_{0,s} = \left(V_s^{\text{slack}}\right)^2, \quad (3.6)$$

where V_s^{slack} is taken directly from the corresponding operating scenario. All bus voltages are constrained by

$$V_{\min}^2 \leq v_{i,s} \leq V_{\max}^2, \quad \forall i \in \mathcal{N}, \quad \forall s \in \Omega_S, \quad (3.7)$$

with $V_{\min} = 0.95$ p.u. and $V_{\max} = 1.05$ p.u.

Each DER's active power output is limited by the available PV generation,

$$0 \leq p_{i,s}^{\text{DER}} \leq p_{i,s}^{\text{PV}}, \quad \forall i \in \Omega_{\text{DER}}, \quad \forall s \in \Omega_S, \quad (3.8)$$

where $p_{i,s}^{\text{PV}}$ denotes the available PV power for DER i in scenario s . In the implementation used in this study, the available PV power is fixed at the rated DER capacity across all scenarios.

This upper bound is further tightened by the local Volt–Watt characteristic, which restricts DER active power output below the available PV generation when the local voltage becomes sufficiently high.

3.2.3 Model Assumptions and Limitations

The OPF formulation relies on several simplifying assumptions that keep the problem computationally tractable while retaining the network behaviour relevant to this study.

The LinDistFlow approximation neglects higher-order loss terms and therefore does not reproduce the full nonlinear AC power-flow solution.

The feeder is modelled as balanced and radial. Three-phase unbalance, neutral conductors and transformer tap-changing are not represented. This keeps the analysis focused on Volt–Watt control without introducing additional network controls.

Only DER active power is controlled. Reactive power support from the DERs is not modelled, although customer reactive demand and the resulting branch reactive power flows remain represented in the network model. As a result, DER-based voltage regulation is achieved through active power curtailment rather than reactive power compensation. Thermal limits are also not imposed, even though branch current ratings are available in the original feeder data. This keeps the analysis focused on voltage-constrained operation, which is directly addressed through Volt–Watt control.

Finally, the OPF is solved over a finite set of operating scenarios. Customer demand and slack-bus voltage vary across these scenarios, while the available PV power is held fixed at the rated DER capacity. The resulting Volt–Watt parameters reflect the range of operating conditions represented by the scenario set. Their performance outside this range is not assessed.

3.3 Local Volt–Watt Control Formulation

The OPF formulation presented above is extended by incorporating the local Volt–Watt characteristic as an additional voltage-dependent constraint on each DER's active power output.

3.3.1 Volt–Watt Constraint

A voltage-dependent upper bound on the active power output of each DER is introduced to represent the Volt–Watt control law. For every DER bus $i \in \Omega_{\text{DER}}$ and operating scenario $s \in \Omega_S$,

$$p_{i,s}^{\text{DER}} \leq p_{i,s}^{\text{PV}} \frac{v_i^{\text{shift}} - v_{i,s}}{V_2^2 - V_1^2}, \quad (3.9)$$

where $p_{i,s}^{\text{PV}}$ is the available photovoltaic generation, $v_{i,s}$ is the squared voltage magnitude at the DER connection point, and v_i^{shift} is the squared-voltage Volt–Watt shift parameter associated with DER i . The constants V_1 and V_2 define the fixed width of the linear curtailment region.

Equation (3.9) is imposed alongside the physical generation limit introduced in Section 3.2 (Equation 3.8), which prevents DER active power output from exceeding the available photovoltaic generation. These constraints determine the feasible active power range of each DER. When the voltage-dependent upper bound exceeds the available photovoltaic generation, the physical generation limit is binding and the DER active power output can reach the available photovoltaic generation. As the local squared voltage increases, the Volt–Watt upper bound decreases linearly until it reaches zero at $v_{i,s} = v_i^{\text{shift}}$.

The Volt–Watt constraint is included directly within the OPF formulation, allowing the network operating points and the DER-specific shift parameters to be determined within the same optimisation problem. The resulting DER active power output is constrained by the corresponding Volt–Watt characteristic across all operating scenarios.

3.3.2 DER-Specific Volt–Watt Shift Parameters

In this study, each DER is assigned a separate Volt–Watt shift parameter, allowing the position of its Volt–Watt characteristic to account for differences in network conditions at different locations along the feeder.

The shift parameter is represented using squared voltage magnitude, consistent with the voltage variables used in the LinDistFlow formulation. For each DER,

$$V_{\text{shift,min}}^2 \leq v_i^{\text{shift}} \leq V_{\text{shift,max}}^2, \quad \forall i \in \Omega_{\text{DER}}, \quad (3.10)$$

where $V_{\text{shift,min}}$ and $V_{\text{shift,max}}$ define the allowable range of the Volt–Watt shift parameter.

A single value of v_i^{shift} is used for each DER across the complete scenario set. Bus voltages, branch power flows and DER outputs vary between scenarios, while the Volt–Watt shift parameters remain fixed. This represents a fixed inverter control setting that is selected to perform across the range of operating conditions included in the scenario set.

At this stage, the three shift parameters v_3^{shift} , v_6^{shift} and v_{12}^{shift} are bounded individually and are otherwise free to take any value within their allowable range.

3.4 Fairness-Constrained Formulation

With the shift parameters independently bounded as described in Section 3.3, the optimisation does not control how feeder capacity is shared among the participating DERs. Since DERs are connected at different electrical locations within the feeder, maximising total DER production alone may result in different utilisation levels between individual DERs.

A fairness-constrained formulation is introduced to reduce these differences. The objective of maximising total DER production remains unchanged. Fairness is incorporated through additional relationships between the DER-specific Volt–Watt shift parameters, which restrict the feasible solution space without introducing an additional term or weighting factor into the objective function.

For a set of participating DERs Ω_{DER} , a fairness-stick coefficient $k_{i,j}$ is introduced for every distinct pair of DER buses $i, j \in \Omega_{\text{DER}}$, with $i < j$. The number of coefficients is

$$\binom{|\Omega_{\text{DER}}|}{2}. \quad (3.11)$$

For the three-DER formulation considered in the main case study, $\Omega_{\text{DER}} = \{3, 6, 12\}$, giving the three coefficients

$$k_{3,6}, \quad k_{3,12}, \quad k_{6,12}, \quad (3.12)$$

which are referred to as the fairness-stick coefficients. Each coefficient satisfies

$$0 < k_{i,j} \leq 1. \quad (3.13)$$

In the numerical implementation, the coefficients are restricted to a narrower search range close to one, since lower values impose increasingly restrictive relationships between the corresponding shift

parameters.

Each coefficient $k_{i,j}$ defines a relative relationship between the shift parameters of a pair of DERs:

$$v_i^{\text{shift}} \leq k_{i,j} v_j^{\text{shift}}, \quad \forall i, j \in \Omega_{\text{DER}}, \quad i < j. \quad (3.14)$$

The convention $i < j$ is used only to define each DER pair once and does not represent an electrical upstream–downstream relationship between the corresponding buses. For the three-DER case, Equation (3.14) gives

$$v_3^{\text{shift}} \leq k_{3,6} v_6^{\text{shift}}, \quad (3.15)$$

$$v_3^{\text{shift}} \leq k_{3,12} v_{12}^{\text{shift}}, \quad (3.16)$$

$$v_6^{\text{shift}} \leq k_{6,12} v_{12}^{\text{shift}}. \quad (3.17)$$

These constraints regulate the relative positions of the DER-specific Volt–Watt characteristics while allowing each DER to retain its own shift parameter. A smaller value of $k_{i,j}$ imposes a stronger relative restriction on v_i^{shift} , while a value closer to one relaxes this restriction.

The fairness-stick coefficients are specified before each OPF evaluation and remain fixed while that OPF is solved. They act as external parameters that determine the feasible relationships between the shift parameters. The shift parameters themselves remain optimisation variables and are solved jointly with the branch power flows, bus voltages and DER active power outputs.

The effect of a candidate set of fairness-stick coefficients is assessed using the average utilisation of each DER. For DER i , this is defined as

$$\bar{u}_i = \frac{1}{|\Omega_S|} \sum_{s \in \Omega_S} \frac{p_{i,s}^{\text{DER}}}{p_{i,s}^{\text{PV}}}. \quad (3.18)$$

The average-utilisation disparity is then defined as

$$D = \max_{i \in \Omega_{\text{DER}}} \bar{u}_i - \min_{i \in \Omega_{\text{DER}}} \bar{u}_i. \quad (3.19)$$

A candidate coefficient set is considered fairness-feasible when

$$D \leq 0.05. \quad (3.20)$$

The 5% disparity limit is therefore applied as an outer candidate-selection criterion, evaluated after each OPF solution rather than imposed as a direct constraint within the OPF itself. Different combinations of fairness-stick coefficients are evaluated according to their resulting total DER production and average-utilisation disparity.

3.5 Optimisation Methods

As discussed in Section 2.5, Bayesian optimisation provides an alternative to exhaustive search when each objective evaluation is computationally expensive. This section describes how grid search and Bayesian optimisation are applied to the fairness-stick coefficients introduced in Section 3.4. Although the optimisation methods are initially applied to the three-DER case, the same search procedures are subsequently extended to the higher-dimensional coefficient spaces considered in the scalability analysis.

For the experiments reported in this study, the number of candidate evaluations is specified before the search begins. Each search completes its prescribed evaluation budget and does not terminate when the first candidate satisfying the 5% disparity limit is identified. After the prescribed budget has been completed, the final solution is selected as the evaluated candidate with the highest total DER production among those satisfying the disparity limit. If no evaluated candidate satisfies the limit, the candidate with the lowest achieved disparity is reported as the best available infeasible solution.

3.5.1 Grid Search

Grid search is applied to the fairness-stick coefficients using a coarse-to-fine strategy. The search first explores the coefficient search space at a coarse resolution, after which a finer local search is performed around the most promising candidate identified during the coarse stage. The grid-search results are then compared with those obtained using Bayesian optimisation.

For the baseline three-DER formulation, the optimisation problem contains three fairness-stick coefficients, $k_{3,6}$, $k_{3,12}$ and $k_{6,12}$. The coarse stage uses four evenly spaced values for each coefficient, giving a complete three-dimensional Cartesian grid of $4^3 = 64$ candidate combinations. A local refinement is then centred on the candidate selected from the coarse stage using a step size of 0.001 over a one-step neighbourhood in each direction. This produces up to a further 27 candidate combinations, with fewer evaluations where clipping at the coefficient bounds produces duplicate points.

As the number of participating DERs increases, the dimensionality of the coefficient space grows according to the pairwise formulation defined in Section 3.4, making complete Cartesian enumeration increasingly difficult within a practical computational budget. For the higher-dimensional formulations investigated in Chapter 4, the coarse-to-fine structure is retained, but only a subset of the coarse Cartesian grid is evaluated. The number of sampled candidates is limited by a predefined evaluation budget, allowing the grid-search benchmark to remain computationally manageable as the dimension of the coefficient space increases. Each candidate evaluation requires solving the complete multi-scenario OPF problem, so the computational cost of grid search depends strongly on the number of coefficient combinations evaluated, particularly as the number of participating DERs increases.

Algorithm 1 Coarse-to-fine grid search for fairness-stick coefficients

- 1: Generate the coarse candidate set using a full Cartesian grid for the three-DER case and sparse sampling for higher-dimensional cases
 - 2: **for** each coarse candidate **do**
 - 3: Solve the multi-scenario OPF and record production and disparity
 - 4: **end for**
 - 5: Select the highest-production feasible candidate, or the lowest-disparity candidate if none is feasible
 - 6: Generate and evaluate a fine local search around the selected candidate
 - 7: Return the highest-production feasible candidate; if none exists, report the lowest-disparity candidate as the best available infeasible candidate
-

3.5.2 Bayesian Optimisation

Bayesian optimisation selects candidate fairness-stick coefficients sequentially using the results of previous OPF evaluations. Instead of evaluating a large number of coefficient combinations according to a predefined grid, the method uses the information obtained from completed evaluations to decide which combination should be evaluated next. This allows the search to concentrate more evaluations in regions that appear promising while still considering regions that have not been explored sufficiently.

The prescribed fairness requirement cannot be evaluated until the corresponding multi-scenario OPF has been solved. Each evaluated coefficient set is therefore assigned a penalised objective value. Candidates satisfying the prescribed disparity limit are evaluated according to their total DER production, while candidates exceeding the limit receive a penalty that increases with the magnitude of the disparity violation. These objective values provide the information used by Bayesian optimisation to guide subsequent evaluations.

The procedure begins with an initial set of coefficient combinations, each of which is evaluated using the multi-scenario OPF. These results are used to construct a Gaussian Process surrogate model, after which further coefficient combinations are selected sequentially using an acquisition function. The final solution is determined from the evaluated OPF results using the selection rule defined at the beginning of Section 3.5. For the higher-dimensional cases investigated in Chapter 4, larger evaluation budgets are used to provide additional search effort as the dimension of the coefficient space increases.

3.5.3 Gaussian Process Surrogate Model

The Gaussian Process (GP) provides an approximate representation of the objective function using the coefficient combinations that have already been evaluated. Based on these known results, it estimates how well an unevaluated coefficient combination is likely to perform without requiring the OPF to be solved at every possible point in the search space.

In addition to estimating performance, the GP indicates the uncertainty associated with each estimate. Regions containing previously evaluated combinations generally have lower uncertainty, while less explored regions have higher uncertainty. This information allows the search to balance the evaluation of promising coefficient combinations with the exploration of less well-characterised regions of the search space.

The GP is used only to support candidate selection. Once a coefficient combination is selected, its actual performance is obtained by solving the multi-scenario OPF. The resulting objective value is then added to the available information and used in the subsequent search.

3.6 Probabilistic Operating Scenario Generation

The proposed optimisation considers 100 equally weighted operating scenarios generated using probabilistic sampling, with each scenario representing one possible operating condition of the distribution feeder. Two uncertain quantities are considered: the slack-bus voltage and the customer load factors at the load buses. The available photovoltaic generation at each DER is held constant across all scenarios and set equal to its installed capacity, allowing the scenario set to represent variations in customer demand and upstream voltage conditions while keeping DER availability unchanged.

Customer load factors are sampled using a correlated truncated normal distribution. A one-factor Gaussian model is used to generate the latent load variables, with a common latent factor introducing positive correlation between the nodal load factors and individual random components allowing demand conditions at different buses to vary between scenarios. The slack-bus voltage is generated using the same latent-variable framework, with a mild negative correlation specified between the feeder-head voltage and aggregate feeder loading to represent lower upstream voltage under higher-loading conditions.

All generated scenarios are required to satisfy the prescribed voltage and load-factor ranges. Samples outside these ranges are discarded and resampled. The latent correlation coefficient is calibrated so that the realised correlation in the final scenario set closely matches the specified target value.

3.7 Model Validation Using an SOC Formulation

As noted in Section 3.2.3, the LinDistFlow approximation neglects higher-order terms in the AC power flow equations. An SOC branch-flow formulation is therefore used to assess the accuracy of the LinDistFlow solutions. The SOC model is used only for validation and does not replace the proposed optimisation model.

The SOC validation uses the same network topology and operating scenarios as the LinDistFlow model. The DER active power outputs obtained from the LinDistFlow optimisation are fixed in the SOC formulation and denoted by $p_{j,s}^{\text{DER, fixed}}$. The SOC model therefore does not determine a new DER dispatch. Instead, bus voltages, branch power flows and branch current magnitudes are recomputed using the fixed DER outputs.

For each non-slack bus j , with $i = \text{parent}(j)$ denoting its upstream bus, the active and reactive power balances are

$$P_{ij,s} = p_{j,s}^L - p_{j,s}^{\text{DER, fixed}} + \sum_{k \in \mathcal{C}(j)} P_{jk,s} + r_{ij} L_{ij,s}, \quad (3.21)$$

and

$$Q_{ij,s} = q_{j,s}^L + \sum_{k \in \mathcal{C}(j)} Q_{jk,s} + x_{ij} L_{ij,s}. \quad (3.22)$$

The corresponding voltage relationship is

$$v_{j,s} = v_{i,s} - 2(r_{ij}P_{ij,s} + x_{ij}Q_{ij,s}) + (r_{ij}^2 + x_{ij}^2)L_{ij,s}, \quad (3.23)$$

where $L_{ij,s}$ denotes the squared branch current magnitude. In contrast to the LinDistFlow approximation, Equation (3.23) retains the second-order term $(r_{ij}^2 + x_{ij}^2)L_{ij,s}$. The active and reactive power balances also include the corresponding branch losses through $r_{ij}L_{ij,s}$ and $x_{ij}L_{ij,s}$.

The relationship between branch power flow, squared current magnitude and upstream squared voltage is represented by the second-order cone constraint

$$P_{ij,s}^2 + Q_{ij,s}^2 \leq v_{i,s}L_{ij,s}. \quad (3.24)$$

With the DER active power outputs fixed, the SOC model is solved by minimising total network losses,

$$\min \sum_{s \in \Omega_S} \sum_{(i,j)} r_{ij}L_{ij,s}. \quad (3.25)$$

The loss-minimisation objective encourages smaller branch current magnitudes while satisfying the SOC branch-flow constraints. Since the DER outputs are fixed, differences between the LinDistFlow and SOC voltage results arise from the different power-flow representations rather than from changes in DER dispatch.

The conic relaxation in Equation (3.24) is not guaranteed to be exact in every scenario. For radial networks, exactness has been established under specific conditions on the underlying optimal power flow problem, including the absence of upper bounds on nodal loads [13]. Since the nodal loads in this study are fixed at their realised scenario values, these theoretical conditions do not directly apply. If the relaxation is not tight, Equation (3.24) may hold as a strict inequality, in which case the resulting solution is feasible for the relaxed SOC model but does not necessarily satisfy the corresponding non-convex branch-flow equality exactly.

For validation, the bus voltages obtained from the SOC formulation are compared with the corresponding LinDistFlow voltages across the same operating scenarios. The feasibility of the SOC solutions and the tightness of the conic relaxation are also examined. These comparisons are reported in Chapter 4 and are used to assess whether the LinDistFlow approximation provides sufficient voltage accuracy for the operating conditions considered in this study.

3.8 Case Study Configuration

The methodology presented in this chapter is demonstrated using the case study configuration summarised in Table 1.

For readability, all voltage parameters in Table 1 are reported as voltage magnitudes in p.u., while their squared values are used where required by the squared-voltage OPF formulation.

The four-, five- and six-DER extensions presented in Section 4.5 use the same network, operating limits, scenario-generation settings and DER rated capacity. The number and locations of the participating DERs are changed, which also changes the number of pairwise fairness-stick coefficients according to the formulation in Section 3.4. The evaluation budgets used for grid search and Bayesian optimisation in these extended cases are reported separately in Chapter 4.

Table 1: Case study configuration for the baseline three-DER formulation

Parameter	Value
Number of buses	13 (radial feeder, Bus 0 slack)
DER locations	Buses 3, 6, 12
DER rated capacity	10 MW each
Voltage limits, $V_{\min}$, $V_{\max}$	0.95, 1.05 p.u.
Volt–Watt reference voltages, V_1 , V_2	1.04, 1.05 p.u.
Volt–Watt shift voltage bounds, $V_{\text{shift},\min}$, $V_{\text{shift},\max}$	1.01, 1.05 p.u.
Fairness-stick coefficient range	0.97 to 1.00
Maximum average-utilisation disparity, D	0.05 (5 percentage points)
Number of operating scenarios	100
Slack-bus voltage range	0.98 to 1.03 p.u.
Load-factor range	0.20 to 0.80
Target nodal load correlation	0.75
Target slack-voltage/aggregate-load correlation	−0.25

3.9 Chapter Summary

This chapter has developed a methodology for determining local Volt–Watt control parameters for DERs connected to a radial low-voltage feeder. A joint OPF model based on the LinDistFlow approximation determines the DER-specific Volt–Watt shift parameters and the corresponding total DER production across a set of probabilistically generated operating scenarios while maintaining the prescribed voltage limits.

For the baseline three-DER formulation, the relationships between the DER-specific Volt–Watt shift parameters are controlled through three pairwise fairness-stick coefficients. Grid search and Bayesian optimisation are used to search for coefficient values that provide high total DER production while satisfying the prescribed 5% average-utilisation disparity limit. The performance of these two search methods is then compared in Chapter 4. The operating scenarios represent variations in customer demand and slack-bus voltage, with correlation introduced between customer load factors and a mild negative correlation imposed between aggregate feeder loading and slack-bus voltage. The LinDistFlow approximation is validated separately using a second-order cone formulation with the DER active power outputs fixed to those obtained from the LinDistFlow solution.

Chapter 4 applies the methodology to the baseline three-DER case and examines the resulting DER production, utilisation and fairness performance. The optimisation framework is then extended to larger DER sets to assess how the computational performance of grid search and Bayesian optimisation changes as the number of fairness-stick coefficients increases.

4 Results

4.1 Overview

This chapter presents the results obtained using the methodology developed in Chapter 3 and the 13-bus low-voltage distribution feeder described in the case study configuration. The unconstrained Volt–Watt formulation is first examined to establish a baseline. The fairness-constrained results are then evaluated by regulating the relative positions of the DER-specific Volt–Watt shift parameters through the fairness-stick coefficients. The resulting changes in DER production, utilisation, curtailment and utilisation disparity are compared with the unconstrained solution.

Grid search and Bayesian optimisation are compared as methods for identifying the fairness-stick coefficients in terms of the number of OPF evaluations, computation time and the quality of the solutions obtained. The computational scalability of the two search methods is then examined by extending the formulation from three to six participating DERs, increasing the number of pairwise fairness-stick coefficients from three to fifteen. Finally, the LinDistFlow-based operating points are validated against an SOC branch-flow formulation to assess the accuracy of the LinDistFlow approximation under the operating conditions considered in this study.

4.2 Baseline Volt–Watt Optimisation

4.2.1 Optimised Parameters and Network Performance

The unconstrained formulation provides the reference case for evaluating the effect of the fairness constraints on Volt–Watt parameter selection. In this case, the OPF model maximises total DER production across all operating scenarios without imposing any constraints on the relative positions of the DER-specific Volt–Watt shift parameters. The three shift parameters are determined independently within their allowable ranges, subject to the network and Volt–Watt constraints.

Table 2 gives the optimised Volt–Watt shift parameters. All three DERs have a shift voltage of 1.05 p.u., corresponding to the maximum allowable value specified in the optimisation model.

Table 2: Optimised Volt–Watt shift voltages obtained for the unconstrained formulation.

DER location	V_{shift} (p.u.)
Node 3	1.0500
Node 6	1.0500
Node 12	1.0500

Figure 2 shows the optimised Volt–Watt characteristics together with the operating points obtained across all operating scenarios.

This result follows directly from the objective of maximising total DER production. A higher shift parameter allows a DER to maintain a higher active power output at higher local voltages. Since the baseline formulation does not include fairness constraints linking the three shift parameters, all three are selected at their upper bound of 1.05 p.u.

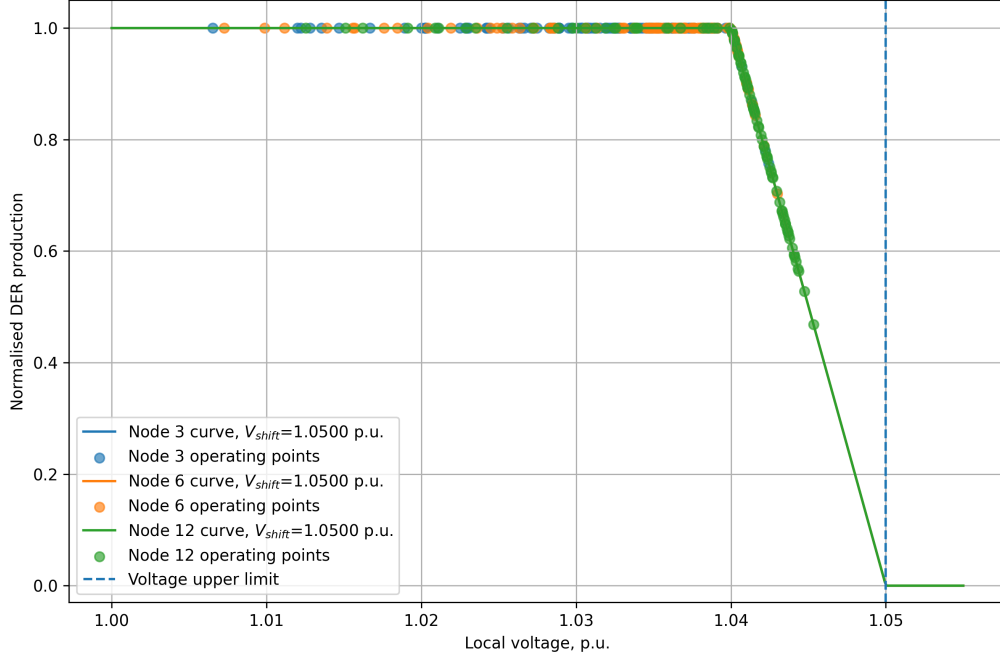


Figure 2: Optimised Volt-Watt characteristics and operating points for the unconstrained formulation.

Across the 100 operating scenarios, the resulting total DER production is 2831.13 MW, corresponding to an average of 28.31 MW per scenario, as shown in Table 3. This is the highest total DER production obtained among the three-DER formulations considered in this study.

Table 3: Overall network performance under the unconstrained formulation.

Performance metric	Value
Number of operating scenarios	100
Total DER production (MW)	2831.13
Average DER production per scenario (MW)	28.31

The unconstrained result serves as the production benchmark for the three-DER case and is used in Section 4.3 to quantify the reduction in total DER production associated with satisfying the prescribed 5% average-utilisation disparity limit.

4.2.2 DER Utilisation and Curtailment

Although all three DERs share the same optimised Volt-Watt shift voltage of 1.05 p.u., their operating performance is different. Table 4 summarises the average utilisation, cumulative curtailment and frequency with which the Volt-Watt upper bound is active for each DER over the complete set of operating scenarios.

Table 4: DER performance under the unconstrained formulation.

DER	Mean utilisation (%)	Total curtailment (MW)	Scenarios with active Volt-Watt constraint (%)
Node 3	99.32	6.84	5
Node 6	98.38	16.25	19
Node 12	85.42	145.78	69

The three DERs achieve different utilisation levels. Nodes 3 and 6 maintain average utilisations of 99.3% and 98.4%, respectively, with very little of their available photovoltaic generation curtailed

under most operating conditions. Node 12 achieves only 85.4%, with a larger share of its available generation curtailed than at Nodes 3 and 6. Since all three DERs use the same optimised Volt–Watt shift voltage, these differences arise from the network conditions experienced at their connection points. Node 12 is located furthest from the slack bus and experiences larger voltage rises during periods of high photovoltaic generation, which results in its Volt–Watt constraint becoming active more frequently and limits its active power output in a larger proportion of the operating scenarios.

The utilisation ordering does not follow a simple hop-count distance from the slack bus. Node 6 is topologically closer to the slack bus, with two branches along its path, than Node 3, with three branches, but exhibits slightly lower utilisation. As shown in Figure 1, the upstream branch between Nodes 1 and 6 supplies a larger downstream subtree than the path towards Node 3. This provides a possible explanation for the slightly lower utilisation observed at Node 6, since local voltage depends on the power flows and impedances along the complete electrical path rather than simply on the number of branches from the slack bus.

Node 12 also experiences the greatest cumulative curtailment, reaching 145.78 MW across the 100 operating scenarios, compared with 16.25 MW for Node 6 and 6.84 MW for Node 3. The Volt–Watt constraint is active in 69% of the operating scenarios at Node 12, compared with 19% at Node 6 and 5% at Node 3.

The importance of electrical location is also evident in the results reported by Olowu et al. [28]. Their Fig. 9(a) shows that PV systems at different electrical locations within the feeder require different optimised Volt–Watt droop characteristics. In their study, this location dependence is reflected in the independently optimised droop curves. In the present study, all three DERs have the same optimised shift voltage, but differences in utilisation and curtailment are still observed. These results show that electrical location can influence DER operating performance even when the same Volt–Watt setting is applied.

4.3 Fairness-Constrained Volt–Watt Optimisation

To address the utilisation differences observed in the unconstrained case, fairness constraints are incorporated into the optimisation framework through the fairness-stick coefficients introduced in Chapter 3. The fairness-stick formulation constrains the relative positions of the DER-specific Volt–Watt shift parameters, with the aim of reducing the utilisation disparity between the participating DERs.

This section examines the effect of the fairness constraints on the optimised Volt–Watt parameters, DER utilisation, active power curtailment, and the resulting production–fairness trade-off.

4.3.1 Optimised Parameters and Network Performance

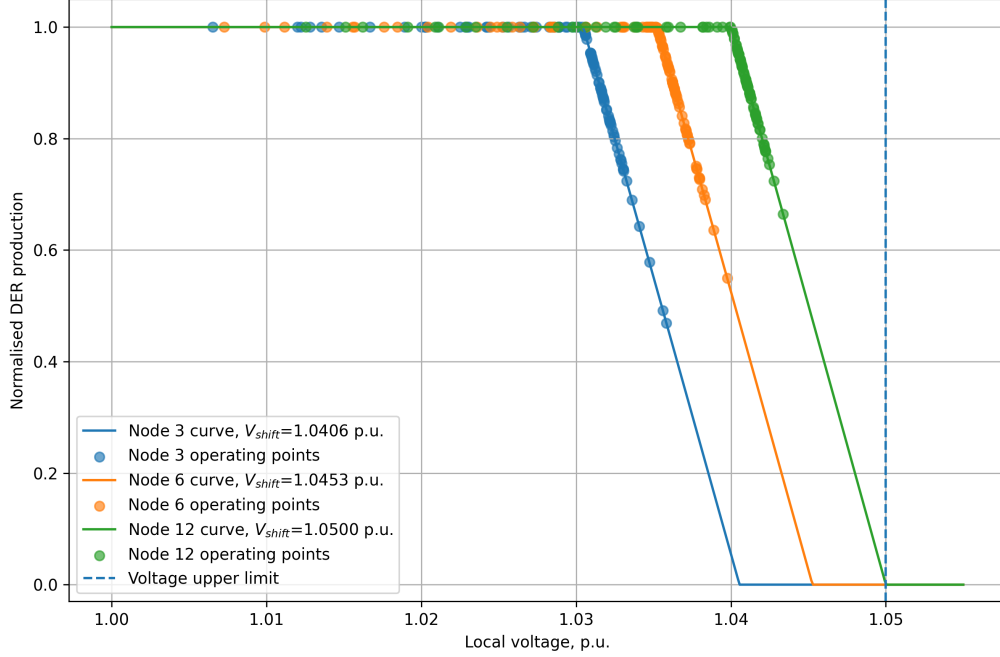
Introducing fairness constraints changes the optimisation problem from independently determining the Volt–Watt shift parameter of each DER to searching for a set of fairness-stick coefficients that regulate the relative positions of the three Volt–Watt characteristics. As described in Chapter 3, grid search and Bayesian optimisation evaluate different combinations of these coefficients and select the feasible candidate with the highest total DER production subject to the prescribed 5% average-utilisation disparity limit.

Despite using different search methods, grid search and Bayesian optimisation produce similar fairness-stick coefficients and resulting Volt–Watt shift voltages, as shown in Table 5.

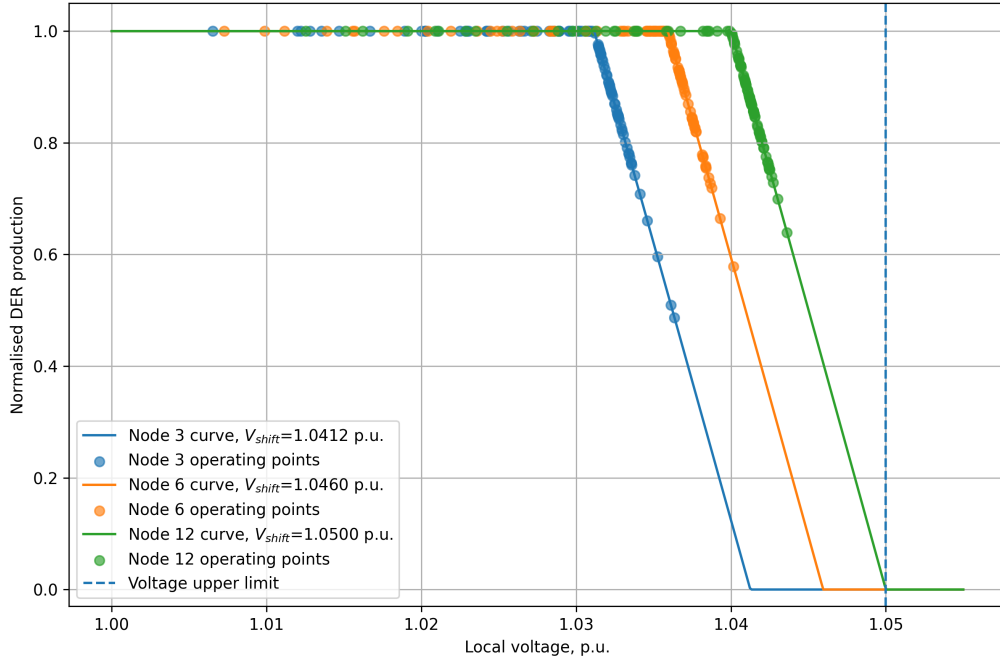
The largest difference between the two solutions occurs for $k_{3,12}$. Despite this difference, the resulting shift voltages are very similar, differing by less than 0.001 p.u. at Nodes 3 and 6 and being identical at Node 12. For both methods, the reported solution is the evaluated candidate with the highest total DER production among those satisfying the 5% disparity limit. The selected coefficients therefore represent the best feasible solution identified by each search, rather than simply the candidate with the lowest utilisation disparity.

Table 5: Optimised fairness-stick coefficients and resulting Volt–Watt shift voltages.

Method	$k_{3,6}$	$k_{3,12}$	$k_{6,12}$	$V_{\text{shift},3}$ (p.u.)	$V_{\text{shift},6}$ (p.u.)	$V_{\text{shift},12}$ (p.u.)
Grid search	0.9910	0.9890	0.9910	1.0406	1.0453	1.0500
Bayesian optimisation	0.9910	0.9851	0.9923	1.0412	1.0460	1.0500



(a) Grid search



(b) Bayesian optimisation

Figure 3: Optimised Volt–Watt characteristics under the fairness-constrained formulation: (a) grid search and (b) Bayesian optimisation.

As shown in Figure 3, the three Volt–Watt curves no longer coincide as they do in the unconstrained

case (Figure 2). Node 3 has the lowest shift voltage, followed by Node 6, while Node 12 remains at the upper bound of 1.05 p.u. The same ordering is obtained by both grid search and Bayesian optimisation, with small differences in the shift voltages reported in Table 5.

Compared with the unconstrained formulation, in which all three shift voltages are 1.05 p.u., the fairness-constrained solutions reduce the shift voltages at Nodes 3 and 6 while maintaining Node 12 at 1.05 p.u. Nodes 3 and 6 therefore become subject to Volt–Watt curtailment at lower local voltages, reducing the utilisation advantage they exhibit in the unconstrained case. Node 12 already operates at the upper bound of the allowable shift-voltage range (Equation 3.10), so its shift voltage cannot be increased further to reduce the utilisation disparity. As a result, the fairness-constrained solution reduces the shift voltages of Nodes 3 and 6, bringing their utilisation closer to that of Node 12.

The resulting network performance is summarised in Table 6.

Table 6: Overall network performance under the fairness-constrained formulations.

Performance metric	Grid search	Bayesian optimisation
Operating scenarios	100	100
Total DER production (MW)	2753.89	2766.36
Average DER production per scenario (MW)	27.54	27.66
Average utilisation disparity (%)	1.26	2.57

Relative to the total DER production of 2831.13 MW in the unconstrained case, production decreases to 2753.89 MW with grid search and 2766.36 MW with Bayesian optimisation, corresponding to reductions of approximately 2.7% and 2.3%, respectively. Average utilisation disparity decreases from 13.9% in the unconstrained formulation to 1.26% with grid search and 2.57% with Bayesian optimisation. Thus, both solutions satisfy the prescribed 5% disparity limit while retaining more than 97% of the total DER production achieved in the unconstrained case.

4.3.2 DER Utilisation and Curtailment

The changes in the Volt–Watt shift parameters presented in Section 4.3.1 directly influence the operating behaviour of the three DERs. Table 7 and Figure 4 compare the average utilisation achieved by each DER under the unconstrained and fairness-constrained formulations.

Table 7: Comparison of average DER utilisation (%).

DER	Unconstrained	Grid search	Bayesian optimisation
Node 3	99.32	91.09	92.12
Node 6	98.38	91.96	93.54
Node 12	85.42	92.34	90.97

The fairness-constrained optimisation reduces the differences in utilisation between the three DERs. Under the unconstrained formulation, Nodes 3 and 6 operate close to their available photovoltaic capacity, while the DER at Node 12 achieves an average utilisation of only 85.4%. After introducing the fairness constraints, the utilisation of Nodes 3 and 6 decreases while that of Node 12 increases. For the grid-search solution, Node 12 reaches an average utilisation of 92.3%, slightly higher than Nodes 3 and 6. Under Bayesian optimisation, Node 12 remains the least-utilised DER at 91.0%. Despite these differences, both solutions satisfy the prescribed 5% average-utilisation disparity limit.

These changes follow directly from the shift-voltage adjustments identified in Section 4.3.1. Under grid search, Node 12’s utilisation increases by 6.9 percentage points relative to the unconstrained case, while those of Nodes 3 and 6 decrease by 8.2 and 6.4 percentage points, respectively. A similar pattern is observed under Bayesian optimisation, with the three DERs operating within a substantially narrower utilisation range than in the unconstrained formulation.

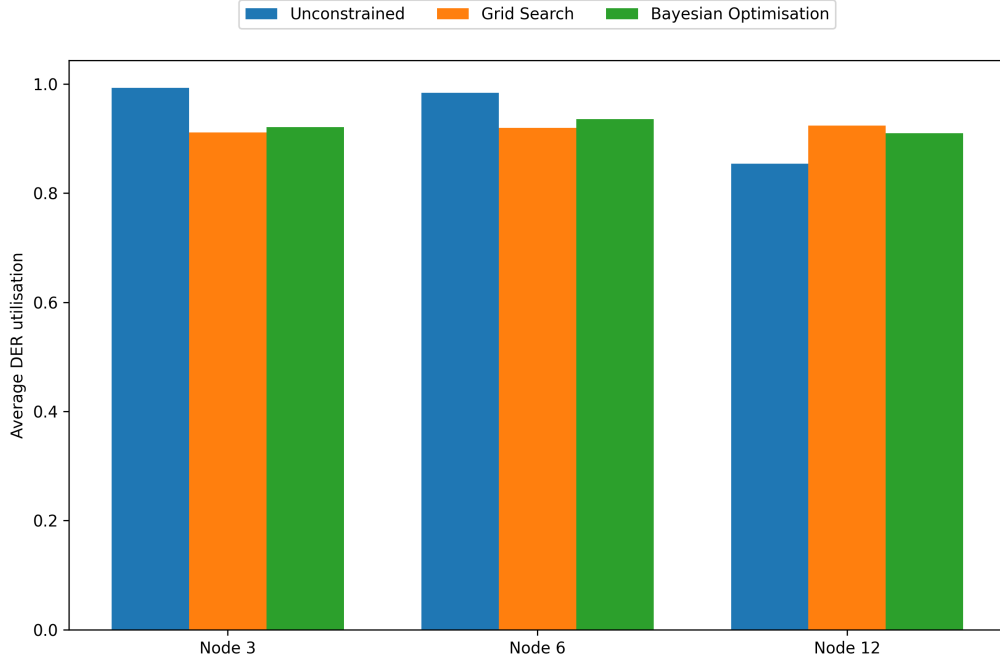


Figure 4: Comparison of average DER utilisation under the unconstrained and fairness-constrained formulations.

Table 8 and Figure 5 compare the cumulative curtailment of each DER under the unconstrained and fairness-constrained formulations.

Table 8: Comparison of cumulative DER curtailment.

DER	Unconstrained (MW)	Grid search (MW)	Bayesian optimisation (MW)
Node 3	6.84	89.13	78.77
Node 6	16.25	80.41	64.59
Node 12	145.78	76.56	90.29

The curtailment results show the corresponding change in the allocation of active power curtailment. Node 12 experiences 145.78 MW of cumulative curtailment in the unconstrained case, which falls to 76.56 MW with grid search and 90.29 MW with Bayesian optimisation. Curtailment at Nodes 3 and 6 increases under both fairness-constrained solutions, reducing the concentration of curtailment at Node 12.

Since the available PV generation is fixed at 10 MW for each DER across all 100 scenarios, the cumulative available generation is 3000 MW over the scenario set. The fairness-constrained solutions redistribute curtailment more evenly among the three DERs, although grid search and Bayesian optimisation produce slightly different allocation patterns because they identify different fairness-stick coefficient combinations.

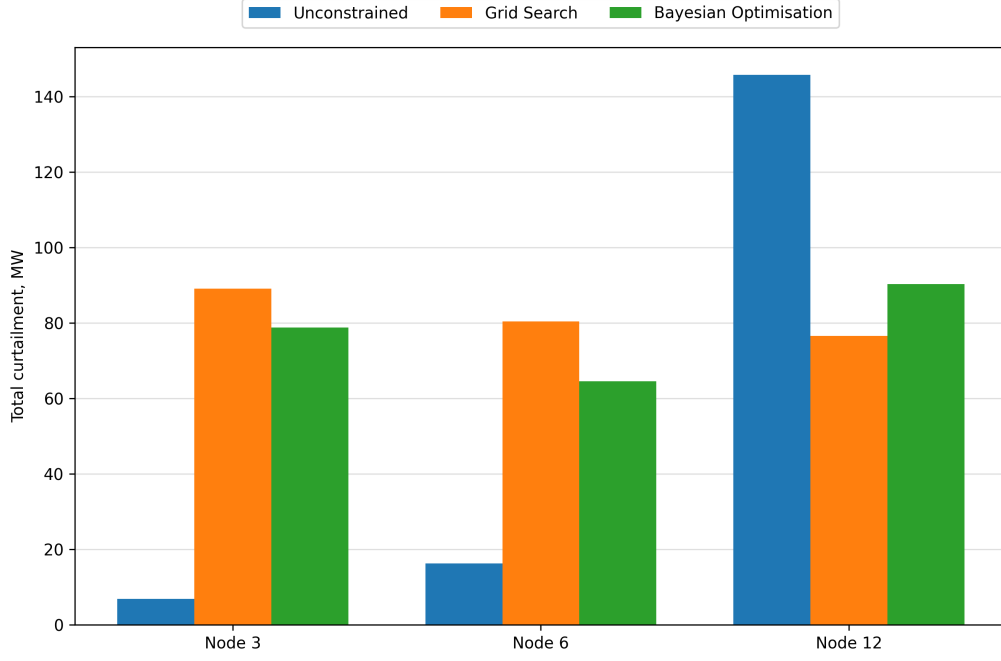


Figure 5: Comparison of cumulative DER curtailment under the unconstrained and fairness-constrained formulations.

4.3.3 Production–Fairness Trade-off

The fairness-constrained formulation involves a trade-off between reducing utilisation disparity and retaining total DER production. Figure 6 shows this relationship across the evaluated fairness-stick coefficient combinations.

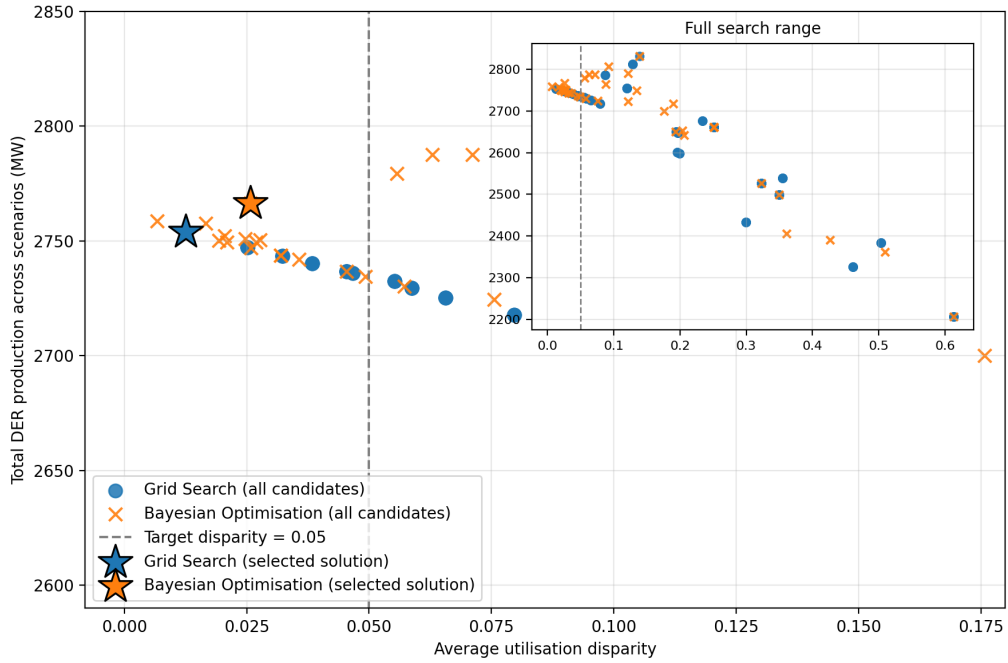


Figure 6: Relationship between total DER production and average utilisation disparity.

For the solutions evaluated in this case study, lower utilisation disparity is accompanied by a reduction in total DER production. The prescribed 5% disparity limit defines the feasible region used to select the final fairness-constrained solution, with each search method selecting the highest-production feasible candidate identified during the search.

This pattern can be explained by the redistribution of curtailment among the three DERs. Bringing the utilisation levels of the three DERs closer together requires additional curtailment at Nodes 3 and 6. Although this is partly offset by the reduction in curtailment at Node 12, the reduction is insufficient to compensate for the additional curtailment introduced at the other two DERs. The resulting increase in total curtailment corresponds to the production reductions of approximately 2.3–2.7% reported in Section 4.3.1.

4.4 Comparison of Search Methods

The previous section compared the operating performance of the fairness-constrained solutions obtained using grid search and Bayesian optimisation. This section compares the two search methods directly in terms of computational cost and solution quality.

4.4.1 Search Efficiency and Solution Quality

Grid search evaluates candidate coefficient combinations using the coarse-to-fine procedure described in Section 3.5.1. For the three-DER case, this consists of a complete coarse Cartesian grid followed by local refinement around the selected coarse candidate. Bayesian optimisation instead selects new candidates sequentially using the results of previous evaluations, as described in Section 3.5.2.

Consistent with the fixed-budget procedure described in Section 3.5, grid search and Bayesian optimisation are assigned 90 and 40 OPF evaluations, respectively, for the three-DER comparison. Table 9 compares the computational requirements and the final fairness-constrained solutions identified by the two methods.

Table 9: Comparison of search efficiency and solution quality.

Metric	Grid search	Bayesian optimisation
OPF evaluations	90	40
Computation time (s)	1363	554
Total DER production (MW)	2753.89	2766.36
Average utilisation disparity (%)	1.26	2.57

Bayesian optimisation requires substantially fewer OPF evaluations and less computation time than grid search while still identifying a solution satisfying the prescribed 5% disparity limit. The two methods locate different feasible points within the parameter space: Bayesian optimisation produces slightly higher total DER production, whereas grid search achieves a lower utilisation disparity. The production difference between the two solutions is small, at approximately 0.45% relative to the grid-search result.

For the three-DER case, Bayesian optimisation therefore identifies a slightly higher-production fairness-feasible solution while using fewer OPF evaluations and less computation time. Grid search produces the lower utilisation disparity, although both solutions remain within the prescribed fairness limit.

4.4.2 Convergence Behaviour

The convergence histories of the two search methods are compared in Figure 7. The curves show the best feasible total DER production identified after each completed OPF evaluation.

Bayesian optimisation identifies its first feasible candidate after 2 evaluations and continues to find improved candidates throughout the search, reaching its final best feasible total DER production of approximately 2766 MW after 36 evaluations. Grid search identifies its first feasible candidate after 10 evaluations but shows little further improvement between evaluations 22 and 66, before reaching its final value of approximately 2754 MW after 84 evaluations.

The horizontal sections of both curves indicate evaluations that do not improve the best feasible solution found at that stage. The convergence histories show that Bayesian optimisation identifies

higher-production feasible candidates earlier in the search and reaches a higher final feasible production using fewer OPF evaluations than grid search.

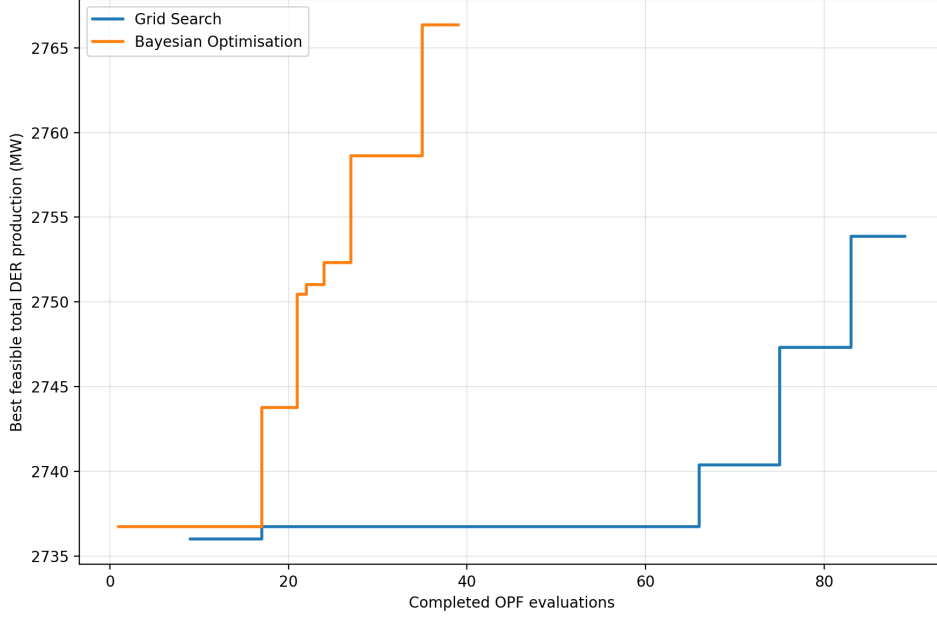


Figure 7: Convergence histories of grid search and Bayesian optimisation.

4.5 Computational Scalability with Increasing Numbers of DERs

4.5.1 Motivation

The original formulation considers three participating DERs. In practical low-voltage distribution networks, larger numbers of DERs may participate in coordinated voltage regulation. As the number of DERs included in the optimisation increases, the dimensionality of the outer optimisation problem also increases because additional pairwise fairness relationships are introduced between the participating DERs.

To investigate the computational scalability of the proposed methodology, three additional case studies are considered by increasing the number of participating DERs from three to four, five and six. The underlying OPF formulation, operating scenarios and Volt–Watt representation remain unchanged throughout these studies. Only the number and locations of the participating DERs and the corresponding fairness-stick coefficients are modified. Following the pairwise formulation defined in Section 3.4, the number of fairness-stick coefficients increases from three in the baseline case to six, ten and fifteen for the four-, five- and six-DER cases, respectively.

These extended cases are used to examine how the computational requirements and solution quality of the two search methods change as the dimension of the fairness-constrained optimisation problem increases.

4.5.2 Optimisation Performance

The extended cases retain the same network and operating-scenario framework while progressively increasing the number of participating DERs. The three-DER configuration includes DERs at Nodes 3, 6 and 12. The four-DER case includes Nodes 3, 6, 9 and 12, the five-DER case includes Nodes 3, 5, 6, 9 and 12, and the six-DER case includes Nodes 3, 5, 6, 9, 11 and 12. Each configuration therefore adds a single DER to the previous case, forming a nested sequence.

The evaluation budgets for the extended cases are specified as experimental computational limits rather than being derived from a convergence criterion. Starting from the original three-DER settings of 90 evaluations for grid search and 40 evaluations for Bayesian optimisation, progressively larger method-specific budgets are assigned as the dimension of the coefficient space increases. Grid search

is allocated 130 and 170 evaluations for the four- and five-DER cases, while Bayesian optimisation is allocated 100 and 160 evaluations, respectively. These values are chosen to provide increasing search effort for each method while keeping the computational requirements manageable. They are not intended to provide an equal-budget comparison between the two methods. For the six-DER case, both methods reach the maximum budget of 200 evaluations adopted for the scalability analysis, chosen to keep the computation time of the largest case at approximately one hour.

The optimisation results obtained under these settings are summarised in Table 10.

Table 10: Comparison of optimisation performance for increasing numbers of participating DERs.

Configuration	Method	Production (MW)	Disparity (%)	Evaluation budget	Target met
3 DER	Grid Search	2753.89	1.26	90	Yes
3 DER	Bayesian Optimisation	2766.36	2.57	40	Yes
4 DER	Grid Search	2828.08	19.49	130	No
4 DER	Bayesian Optimisation	2871.51	2.21	100	Yes
5 DER	Grid Search	3006.01	22.84	170	No
5 DER	Bayesian Optimisation	3157.67	4.99	160	Yes
6 DER	Grid Search	2785.10	48.12	200	No
6 DER	Bayesian Optimisation	3080.03	2.98	200	Yes

Grid search satisfies the prescribed 5% disparity limit for the three-DER formulation, but does not identify a fairness-feasible solution for the four-, five- or six-DER cases within the specified budgets. Bayesian optimisation satisfies the disparity limit in all four configurations. The five-DER Bayesian-optimisation result of 4.99% lies close to the prescribed 5% disparity limit, indicating only a small feasibility margin for this configuration. In the six-DER case, where both methods use the same budget of 200 evaluations, Bayesian optimisation achieves a disparity of 2.98%, while the grid-search result remains above the prescribed limit at 48.12%.

The effect of the fairness requirement on total DER production is summarised in Table 11.

Table 11: Reduction in total DER production associated with the fairness constraint.

Configuration	Unconstrained	Fairness-constrained	Reduction (%)
3 DER	2831.13	2766.36	2.29
4 DER	3070.90	2871.51	6.49
5 DER	3519.56	3157.67	10.28
6 DER	3542.19	3080.03	13.05

For the four-, five- and six-DER formulations, the fairness-constrained production reported in Table 11 corresponds to the Bayesian optimisation solution because grid search does not identify a solution satisfying the 5% disparity limit in these cases. The Bayesian optimisation result is also used for the three-DER case so that the production reductions across all four configurations are calculated using solutions obtained by the same search method.

The unconstrained production also increases by only 0.6% from the five-DER to the six-DER configuration, despite the addition of a further 10 MW DER at Node 11. This indicates that, under the network conditions considered, the additional installed capacity provides little net increase in DER production. Under the fairness constraint, the change is more pronounced. The total DER production of the Bayesian-optimisation solution decreases from 3157.67 MW in the five-DER case to 3080.03 MW in the six-DER case, indicating that, for these configurations, the additional curtailment associated with satisfying the 5% disparity limit outweighs the production increase obtained from the additional DER.

Across the four configurations, the reduction in total DER production relative to the unconstrained solution increases from 2.29% for three DERs to 6.49%, 10.28% and 13.05% for four, five and six DERs, respectively. For the configurations examined in this case study, maintaining the prescribed fairness level among a larger number of participating DERs therefore requires a greater sacrifice in total DER production. This reflects the increasing amount of curtailment required to keep utilisation disparity within the 5% limit as fairness is enforced across more DER locations.

4.5.3 Computational Performance

For optimisation problems beyond the original three-DER formulation, the grid-search benchmark follows the coarse-to-fine procedure described in Chapter 3, evaluating only a subset of the candidate combinations as the number of fairness-stick coefficients increases.

Table 12 compares the total computation time of the two search methods across the four DER configurations. The reported runtimes correspond to completion of the full prescribed evaluation budget for each method.

Table 12: Comparison of total computation time for increasing numbers of participating DERs.

Method	3 DER	4 DER	5 DER	6 DER
Grid Search (s)	1363	2526	3284	3909
Bayesian Optimisation (s)	554	2042	3047	3556

Computation time increases for both methods as the number of participating DERs and the evaluation budgets reported in Table 10 increase. Because the two methods use different evaluation budgets in the three-, four- and five-DER cases, the corresponding runtimes reflect both the number of OPF evaluations and the computational requirements of the search procedures. These cases should therefore not be interpreted as equal-budget timing comparisons. The six-DER case provides the direct comparison, with both methods completing 200 evaluations and Bayesian optimisation requiring less computation time than grid search under this common budget.

The total runtimes in Table 12 describe the time required to complete the full search budget. For Bayesian optimisation, the completed search histories can also be examined retrospectively to determine when the final reported fairness-feasible solution was first encountered. Table 13 reports this evaluation together with the cumulative computation time up to that point. These cumulative times are distinct from the total Bayesian-optimisation runtimes reported in Table 12 and do not represent actual stopping times.

Table 13: Retrospective identification of the final Bayesian-optimisation solution within the prescribed evaluation budget.

Configuration	Evaluation budget	Final solution first identified at	Cumulative time at first identification (s)
4 DER	100	38	754
5 DER	160	147	2794
6 DER	200	95	1770

These identification points are determined retrospectively from the completed search histories, so they cannot be interpreted as stopping points available during the optimisation. At the time each solution was first encountered, it was not yet known whether a better fairness-feasible solution would be identified by a later evaluation. The results nevertheless show that the evaluation at which the final selected solution first appears does not increase monotonically with the number of participating DERs. In particular, the five-DER case requires 147 evaluations before reaching its final reported solution, while the six-DER case reaches its final solution at evaluation 95 despite having a larger coefficient space. The five-DER result also shows the importance of allowing a sufficient evaluation

budget in the higher-dimensional searches, since its final selected solution is encountered only near the end of the prescribed budget.

Grid search evaluates candidates according to a sampling procedure defined before the corresponding OPF results are known. As the number of fairness-stick coefficients increases, a given evaluation budget covers a smaller proportion of the coefficient search space. The proportion of infeasible OPF evaluations is approximately 4.6%, 14.1% and 22.0% for the four-, five- and six-DER cases, respectively. These evaluations consume part of the available computational budget without providing feasible candidate solutions.

Bayesian optimisation uses the results of completed evaluations to select subsequent coefficient combinations. Its proportion of infeasible OPF evaluations is approximately 4.0%, 11.2% and 27.0% for the four-, five- and six-DER cases, respectively. In the six-DER case, this proportion is higher than that observed for grid search. One possible explanation is the increasing dimensionality of the coefficient space, since the number of fairness-stick coefficients increases to fifteen while the available evaluation budget remains limited to 200 evaluations. This results in increasingly sparse sampling of the higher-dimensional search space and may reduce the ability of the surrogate model to distinguish feasible and infeasible regions. However, the results obtained here are not sufficient to establish this as the cause of the higher infeasible fraction.

The computational effort required by Bayesian optimisation also increases as the coefficient space grows. Its runtime increases from the three-DER to the six-DER formulation, showing that Bayesian optimisation does not remove the computational cost associated with increasing problem dimension. Within the budgets considered, however, it continues to identify fairness-feasible solutions as the number of participating DERs increases, whereas the grid-search benchmark does not identify such a solution beyond the three-DER case.

4.6 SOC Validation

To evaluate the accuracy of the LinDistFlow formulation, the operating points obtained from each optimisation formulation are re-evaluated using the second-order cone (SOC) branch-flow model described in Chapter 3. The SOC model does not determine new DER operating points. Instead, it fixes the DER active power outputs obtained from the LinDistFlow optimisation and recomputes the resulting network voltages. This comparison assesses the accuracy of the linearised network model under the operating conditions considered in this study.

4.6.1 Voltage Accuracy

Voltage accuracy is assessed by comparing the bus voltages obtained from the LinDistFlow formulation with those recomputed using the SOC model. For each operating scenario, the maximum absolute voltage difference across all buses is calculated. The resulting errors for the three optimisation formulations are summarised in Table 14.

Table 14: Voltage accuracy obtained from SOC validation.

Formulation	Mean scenario maximum voltage error (p.u.)	Overall maximum voltage error (p.u.)
Unconstrained	0.001324	0.001700
Fairness-constrained (Grid Search)	0.001319	0.001700
Fairness-constrained (Bayesian Optimisation)	0.001318	0.001700

The mean scenario maximum voltage error is approximately 0.00132 p.u. for all three formulations. The overall maximum error across all buses and operating scenarios is 0.00170 p.u. in each case. The similarity of these values shows that the LinDistFlow approximation produces comparable voltage accuracy for the three sets of operating points considered.

The overall maximum voltage difference between the LinDistFlow and SOC results is equivalent to 0.17% of nominal voltage. The SOC comparison therefore indicates that the LinDistFlow formulation provides a close approximation of the bus voltages obtained from the SOC-relaxed model for the network and operating scenarios considered in this study.

4.6.2 Feasibility and Relaxation Tightness

The SOC validation problem is successfully solved for all 100 operating scenarios under each optimisation formulation, and no voltage violations are observed. This confirms that the fixed DER operating points obtained from the LinDistFlow optimisation produce feasible solutions to the SOC-relaxed model within the prescribed voltage limits. Table 15 summarises these results.

Table 15: SOC feasibility and relaxation tightness.

Formulation	Validated scenarios	Voltage violations	Scenarios satisfying relaxation tightness criterion (%) (maximum cone gap $\leq 10^{-5}$)
Unconstrained	100	0	44
Fairness-constrained (Grid Search)	100	0	40
Fairness-constrained (Bayesian Optimisation)	100	0	37

The relaxation tightness percentage represents the percentage of the 100 validated scenarios in which the maximum absolute cone gap across all branches does not exceed the numerical tolerance of 10^{-5} . Across the three formulations, between 37% and 44% of the scenarios satisfy this criterion.

For the scenarios classified as tight, the second-order cone constraint in Equation (3.24) is satisfied at equality to within the prescribed numerical tolerance. In the remaining scenarios, the maximum cone gap exceeds 10^{-5} , so the SOC relaxation is not classified as tight and is not guaranteed to coincide with a solution of the original non-convex branch-flow equations. As discussed in Section 3.7, theoretical exactness results for radial networks require specific conditions that are not directly satisfied by the validation problem considered here [13].

Given this limitation, the voltage errors reported in Table 14 measure the difference between the LinDistFlow voltages and those obtained from the SOC-relaxed model, rather than the error relative to the full non-convex AC power-flow equations in every scenario. The SOC formulation nevertheless retains the second-order loss term omitted by LinDistFlow and provides a more detailed representation of branch losses and voltage drops for the comparison.

The voltage errors are also compared between scenarios in which the SOC relaxation is tight and those in which it is not. Across the three formulations, the mean scenario maximum voltage error in the non-tight scenarios is only 0.00014–0.00018 p.u. higher than in the tight scenarios. For the unconstrained formulation, for example, the corresponding values are approximately 0.00139 p.u. and 0.00124 p.u. The largest errors observed in the two groups are also similar, at approximately 0.00170 p.u. for the non-tight scenarios and 0.00169 p.u. for the tight scenarios.

The similar voltage-error magnitudes observed in the two groups indicate that the non-tight scenarios do not exhibit a marked increase in the discrepancy between the LinDistFlow and SOC-relaxed voltage results. Together with the small voltage differences reported in Section 4.6.1, these results support the use of LinDistFlow as a sufficiently accurate representation of bus voltages for the feeder and operating scenarios considered in this study. However, the limited tightness of the SOC relaxation means that this validation should not be interpreted as an exact comparison with the full non-convex AC power-flow equations in every scenario. The same level of voltage accuracy should therefore not be assumed for networks with different impedance characteristics, topologies or operating conditions without further validation.

4.7 Chapter Summary

The unconstrained Volt–Watt formulation achieves the highest total DER production in the three-DER case, but curtailment is unevenly distributed across the feeder, with Node 12 experiencing the largest reduction in utilisation. Introducing the fairness-stick coefficients changes the distribution of curtailment across the three DERs and reduces the average utilisation disparity. Both fairness-constrained solutions satisfy the prescribed 5% disparity limit while retaining more than 97% of the total DER production achieved in the unconstrained case.

For the three-DER formulation, grid search and Bayesian optimisation both identify solutions satisfying the fairness requirement, while Bayesian optimisation uses fewer OPF evaluations and achieves slightly higher total DER production. When the formulation is extended to four, five and six DERs, the difference between the two search methods becomes clearer. Under the evaluation budgets used for each configuration, Bayesian optimisation identifies a solution satisfying the 5% disparity limit in every case, whereas grid search does not identify a fairness-feasible solution beyond the three-DER formulation. The computational requirements of both methods increase as the number of participating DERs and fairness-stick coefficients increases.

The SOC validation shows small voltage differences between the LinDistFlow and SOC-relaxed models for the operating scenarios considered. The SOC relaxation is not tight in every scenario, so this comparison cannot be interpreted as an exact validation against the full non-convex AC power-flow equations in all cases. The results nevertheless support the use of the LinDistFlow approximation for representing bus voltages within the optimisation framework considered in this study.

5 Conclusions and Future Work

5.1 Conclusions

This dissertation set out to determine Volt–Watt control parameters that maximise total DER production in a low-voltage distribution network while limiting the disparity in utilisation among participating DERs. The three objectives stated in Section 1.2 were addressed through the development and validation of the OPF-based formulation, the application of Bayesian optimisation to the fairness-constrained parameter search, and its comparison with grid search across different DER configurations.

The OPF-based formulation developed in Chapter 3 provides a means of determining DER-specific Volt–Watt parameters across multiple operating scenarios while accounting for network voltage limits and DER curtailment. In the unconstrained three-DER case, maximising total DER production results in an uneven distribution of curtailment, with the downstream DER at Node 12 achieving lower utilisation than the other participating DERs. Comparison with the SOC-relaxed branch-flow model shows that the LinDistFlow formulation produces closely matching network voltages under the operating conditions considered, although the conic relaxation is not tight in every scenario and this comparison should not be interpreted as an exact validation against the full non-convex power-flow equations.

Fairness is introduced through the fairness-stick coefficients, which regulate the relative positions of the DER-specific Volt–Watt shift parameters. In the three-DER case, both search methods substantially reduce the average utilisation disparity from 13.9% in the unconstrained formulation, achieving 1.26% with grid search and 2.57% with Bayesian optimisation. Both retain more than 97% of the unconstrained total DER production. The resulting shift parameters increase curtailment at Nodes 3 and 6 and reduce the concentration of curtailment at Node 12, producing a more balanced distribution of DER utilisation.

The comparison of search methods shows a different pattern as the coefficient space increases in dimension. For the three-DER case, grid search and Bayesian optimisation identify fairness-constrained solutions of comparable quality, while Bayesian optimisation requires less than half as many OPF evaluations and less than half the computation time. For the four-, five- and six-DER cases, Bayesian optimisation continues to identify solutions satisfying the prescribed 5% disparity limit within the evaluation budgets used. Grid search does not identify a fairness-feasible solution for these higher-dimensional cases within the corresponding budgets.

Across the configurations investigated, the Bayesian-optimised solutions satisfy the prescribed 5% average-utilisation disparity limit in every case. The reduction in total DER production associated with the fairness constraint increases from 2.29% for the three-DER case to 6.49%, 10.28% and 13.05% for the four-, five- and six-DER cases, respectively. Within this case study, the results therefore show an increasing production cost of fairness as the number of participating DERs increases.

These trends also have implications for the choice of search method. As the coefficient space grows, grid search covers an increasingly small proportion of the possible coefficient combinations for a given evaluation budget, making fairness-feasible regions more difficult to identify within a practical computational budget. Bayesian optimisation instead uses information from previous evaluations to direct subsequent searches towards more promising regions of the coefficient space. The results obtained in the extended cases point to the value of an adaptive search strategy when coordinated Volt–Watt parameter selection is applied to larger numbers of DERs.

5.2 Limitations and Future Work

The results of this dissertation should be interpreted within the modelling assumptions and experimental conditions considered.

The network model assumes a balanced radial feeder and uses a LinDistFlow approximation, with the voltage results validated against an SOC branch-flow formulation. This representation is suitable for the feeder considered in this study, but practical low-voltage networks may exhibit three-phase unbalance and more complex network structures. Future work could extend the framework to unbalanced three-phase models and assess whether the resulting Volt–Watt settings and fairness outcomes remain consistent under a wider range of network conditions.

A more fundamental limitation arises from the pairwise fairness-stick parameterisation. Increasing the number of participating DERs from three to six increases the number of fairness-stick coefficients from three to fifteen. The resulting growth in dimensionality increases the size of the search space and makes parameter selection more computationally demanding. The search-method comparison in this study is limited to configurations of up to six DERs and fifteen fairness-stick coefficients. Whether the observed benefit of Bayesian optimisation within the prescribed evaluation budgets persists for substantially larger DER configurations has not been assessed and requires further investigation.

The case study also adopts several controlled experimental assumptions, including equal DER capacities and a fixed set of 100 operating scenarios. These choices provide a consistent basis for comparing the search methods, but they do not represent the full range of DER sizes, operating conditions and stochastic variation that may occur in practice. Further studies could examine heterogeneous DER capacities and locations, alternative operating-scenario sets, and the sensitivity of the optimisation results to different scenario realisations.

The present study focuses on active power control through Volt–Watt characteristics. Reactive power support is outside the scope of the analysis, but coordinated Volt–Var and Volt–Watt control provides a natural extension. Such an extension could examine whether reactive power support reduces the amount of active power curtailment required to maintain voltage limits and how this affects DER utilisation fairness.

References

- [1] M. A. A. Al-Jaafreh and G. Mokryani. Planning and operation of LV distribution networks: A comprehensive review. *IET Energy Systems Integration*, 1(3):133–146, 2019.
- [2] O. A. Balogun, Y. Sun, and P. A. Gbadega. Optimal PV active power curtailment in a PV-penetrated distribution network using optimal smart inverter Volt-Watt control settings. *Energy Reports*, 12:5396–5419, 2024.
- [3] M. E. Baran and F. F. Wu. Network reconfiguration in distribution systems for loss reduction and load balancing. *IEEE Transactions on Power Delivery*, 4(2):1401–1407, 1989.
- [4] J. Bergstra and Y. Bengio. Random search for hyper-parameter optimization. *Journal of Machine Learning Research*, 13(2), 2012.
- [5] B. Bischl, M. Binder, M. Lang, et al. Hyperparameter optimization: Foundations, algorithms, best practices, and open challenges. *Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery*, 13(2):e1484, 2023.
- [6] M. H. J. Bollen, Y. Yang, and F. Hassan. Integration of distributed generation in the power system - a power quality approach. In *2008 13th International Conference on Harmonics and Quality of Power*, pages 1–8. IEEE, 2008.
- [7] E. Brochu, V. M. Cora, and N. De Freitas. A tutorial on Bayesian optimization of expensive cost functions, with application to active user modeling and hierarchical reinforcement learning. *arXiv preprint arXiv:1012.2599*, 2010.
- [8] A. da Costa Lopes, L. E. S. e Silva, J. P. A. Vieira, et al. An enhanced voltage control method for fair active power curtailment of rooftop PV systems in LV distribution networks. *Sustainable Energy, Grids and Networks*, 39:101409, 2024.
- [9] E. Dall’Anese and A. Simonetto. Optimal power flow pursuit. *IEEE Transactions on Smart Grid*, 9(2):942–952, 2016.
- [10] N. Damianakis, G. R. C. Mouli, and P. Bauer. Grid impact of photovoltaics, electric vehicles and heat pumps on distribution grids – an overview. *Applied Energy*, 380:125000, 2025.
- [11] E. Demirok, P. C. Gonzalez, K. H. B. Frederiksen, et al. Local reactive power control methods for overvoltage prevention of distributed solar inverters in low-voltage grids. *IEEE Journal of Photovoltaics*, 1(2):174–182, 2011.
- [12] M. Emmanuel, J. Giraldez, P. Gotseff, et al. Estimation of solar photovoltaic energy curtailment due to Volt-Watt control. *IET Renewable Power Generation*, 14(4):640–646, 2020.
- [13] M. Farivar and S. H. Low. Branch flow model: Relaxations and convexification – Part I. *IEEE Transactions on Power Systems*, 28(3):2554–2564, 2013.
- [14] S. Frank and S. Rebennack. An introduction to optimal power flow: Theory, formulation, and examples. *IEEE Transactions*, 48(12):1172–1197, 2016.
- [15] D. Gebbran, S. Mhanna, Y. Ma, et al. Fair coordination of distributed energy resources with Volt-Var control and PV curtailment. *Applied Energy*, 286:116546, 2021.
- [16] Y. Z. Gerdroodbari, R. Razzaghi, and F. Shahnia. Decentralized control strategy to improve fairness in active power curtailment of PV inverters in low-voltage distribution networks. *IEEE Transactions on Sustainable Energy*, 12(4):2282–2292, 2021.
- [17] Y. Gui, K. Nainar, C. I. Ciontea, et al. Automatic voltage regulation application for PV inverters in low-voltage distribution grids – a digital twin approach. *International Journal of Electrical Power & Energy Systems*, 149:109022, 2023.

- [18] J. M. Hernández-Lobato, M. A. Gelbart, R. P. Adams, et al. A general framework for constrained Bayesian optimization using information-based search. *Journal of Machine Learning Research*, 17(160):1–53, 2016.
- [19] IEEE. IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces. *IEEE Std 1547-2018*, pages 1–138, 2018.
- [20] A. Inaolaji, A. Savasci, and S. Paudyal. Optimal droop settings of smart inverters. In *2021 IEEE 48th Photovoltaic Specialists Conference (PVSC)*, pages 2584–2589. IEEE, 2021.
- [21] International Energy Agency. Renewables 2025: Analysis and forecasts to 2030. Technical report, IEA, Paris, 2025.
- [22] M. Kang, G. Yoon, H. Jin, et al. Domain-based local fairness control guided by voltage sensitivities for high-PV distribution feeders. *Sustainable Energy, Grids and Networks*, 47:102335, 2026.
- [23] M. G. Kashani, M. Mobarrez, and S. Bhattacharya. Smart inverter Volt-Watt control design in high PV-penetrated distribution systems. *IEEE Transactions on Industry Applications*, 55(2):1147–1156, 2018.
- [24] H. A. Khan, M. Zuhaib, and M. Rihan. Voltage fluctuation mitigation with coordinated OLTC and energy storage control in high PV penetrating distribution network. *Electric Power Systems Research*, 208:107924, 2022.
- [25] J. Licari, S. E. Rhaili, A. Micallef, et al. Addressing voltage regulation challenges in low voltage distribution networks with high renewable energy and electrical vehicles: A critical review. *Energy Reports*, 14:2977–2997, 2025.
- [26] M. Z. Liu, A. T. Procopiou, K. Petrou, et al. On the fairness of PV curtailment schemes in residential distribution networks. *IEEE Transactions on Smart Grid*, 11(5):4502–4512, 2020.
- [27] K. Luo and W. Shi. Comparison of voltage control by inverters for improving the PV penetration in low voltage networks. *IEEE Access*, 8:161488–161497, 2020.
- [28] T. O. Olowu, A. Inaolaji, A. Sarwat, et al. Optimal Volt-Var and Volt-Watt droop settings of smart inverters. In *2021 IEEE Green Technologies Conference (GreenTech)*, pages 89–96. IEEE, 2021.
- [29] T. Rahimi, S. Ahmed, J. L. Cardenas-Barrera, et al. Fairness-oriented Volt–Watt control methods of PV units for over-voltage suppression in PV-enriched smart cities. *Smart Cities*, 8(3):88, 2025.
- [30] H. Sekhavatmanesh, G. Ferrari-Trecate, and S. Mastellone. Optimal adaptive droop design via a modified relaxation of the OPF. *IEEE Transactions on Control Systems Technology*, 31(2):497–510, 2022.
- [31] D. Stanelytė and V. Radziukynas. Analysis of voltage and reactive power algorithms in low voltage networks. *Energies*, 15(5):1843, 2022.
- [32] D. Stenger, M. Nitsch, and D. Abel. Joint constrained Bayesian optimization of planning, guidance, control, and state estimation of an autonomous underwater vehicle. In *2022 European Control Conference (ECC)*, pages 1982–1987. IEEE, 2022.
- [33] R. Tonkoski, L. A. C. Lopes, and T. H. M. El-Fouly. Coordinated active power curtailment of grid connected PV inverters for overvoltage prevention. *IEEE Transactions on Sustainable Energy*, 2(2):139–147, 2011.
- [34] R. Wagle, P. Sharma, C. Sharma, et al. Optimal power flow based coordinated reactive and active power control to mitigate voltage violations in smart inverter enriched distribution network. *International Journal of Green Energy*, 21(2):359–375, 2024.

- [35] K. Yamane, D. Orihara, D. Iio, et al. Determination method of Volt-Var and Volt-Watt curve for smart inverters applying optimization of active/reactive power allocation for each inverter. *Electrical Engineering in Japan*, 209(1-2):10–19, 2019.
- [36] S. Yoshizawa, Y. Yanagiya, H. Ishii, et al. Voltage-sensitivity-based Volt-Var-Watt settings of smart inverters for mitigating voltage rise in distribution systems. *IEEE Open Access Journal of Power and Energy*, 8:584–595, 2021.